

CQUniversity Australia

CQU Rockhampton TAFE Consolidation – Stage 2

VERTICAL TRANSPORTATION SPECIFICATION

FEBRUARY 2026



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CQU Rockhampton TAFE CONSOLIDATION – STAGE 2

Vertical Transportation Specification

CQUniversity Australia

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1 PROJECT INFORMATION

1.1 GENERAL

WSP Australia Pty Limited WSP has been engaged for the CQU TAFE Redevelopment Project as part of design team to deliver these specifications considering existing and future demands. Based on a 2022 masterplan study, the scope includes the provision of a new Trades Building (Building 92), and the refurbishment of Buildings 19, 33, 34, 36 and Engineering & Horticulture.

As part of the initial scope of services, WSP attended the Rockhampton North and City Campuses to undertake a high-level review of the existing Electrical Services infrastructure reticulating around each site.

This specification takes into consideration existing items of plant and equipment, and its capacity for the proposed expansion.

The project delivery being design and construct, has been adopted for this project.

This Specification is to be read in conjunction with other tender documents issued by the Project Manager including:

- Principal's Project Requirements
- Conditions of Tender
- Conditions of Contract
- Contract Preliminaries

This Specification is a technical specification and for the purpose of this Specification, Works shall mean all works or work as described in this Specification and/or associated drawings

This technical specification and drawings are intended for a Design and Construct Contract and have been provided to communicate the design intent of the Works for the project and for the guidance of tenderers pricing the project. Notwithstanding this, the completion of the design, the co-ordination of the electrical services works, and achieving the system performance covered by this technical specification & Drawings shall form part of the Works.

The drawings and scope of works are based on information and drawings provided by the Client, other stakeholders and other design consultants which may contain conflicting data, incomplete design, coordination, or omissions. These layouts will be subject to change to suit final design. The contractor pricing the works shall make an assessment and make adequate contingency based on their experience with the procurement and risks involved in projects of this nature i.e., changes to requirements, site investigation outcomes, scope, and future design detail.

The Works requires the necessary resources to complete the detailed "For Construction" documentation and schedules, final equipment selections, testing and commissioning of the installation in accordance with the design intent and the requirements of this technical specification and drawings.

Notwithstanding this, achieving the system performance covered by this technical specification & drawings shall form part of the Works.

The work shall comprise that which is described in this document, any subsequent addendums.

The Drawings shall be read in conjunction with the respective existing services drawings and manuals and all associated documentation.

The tenderer shall visit the site to fully familiarise themselves with existing services installation and site conditions. Allowance shall be made to coordinate the existing services with the project staging and the scope of works covered under this Contract.

Include for all applicable site allowances and overtime required to meet the project programme.

Tender prices are to be submitted as a lump sum, fixed for the duration of the contract.

1.1.1 ASSOCIATED DOCUMENTS

This technical specification is to be read in conjunction with the project documents and it shall be taken that for the purpose of this specification and scope of works all associated drawings and technical specifications including but not limited to those below have been received, perused, and acknowledged in the submission of a commercial offer for the scope of works:

- 1 Site Survey Drawings
- 2 Architectural and Interior Design
- 3 Structural & Civil Drawings
- 4 Electrical Services
- 5 Mechanical Services
- 6 Fire Protection Services
- 7 Hydraulic Services
- 8 Fire Engineering Report

1.2 PROJECT TEAM

The Project Team includes the following team members:

PRINCIPAL/CLIENT	CQU
SUPERINTENDENT	TBC
PROJECT MANAGER	TBC
CONTRACTOR	The company who undertakes the Works
ARCHITECT	Thompson Adsett & Peddle Thorp
SERVICES CONSULTANT ENGINEER (MECHANICAL, ELECTRICAL, FIRE PROTECTION, HYDRAULICS)	WSP Australia Pty Limited
CIVIL ENGINEER	WSP Australia Pty Limited

STRUCTURAL ENGINEER	WSP Australia Pty Limited
FIRE SAFETY ENGINEER	WSP Australia Pty Limited
BUILDING SURVEYOR	TBC

1.3 THE PROJECT DESCRIPTION

1.3.1 GENERAL DESCRIPTION

The project is the consolidation of the existing Rockhampton CQU and TAFE site. The project comprises of:

- A new Trades Centre Building
- Refurbishment Building 19 level-3
- Partial/small refurbishment of building 32
- Refurbishment Building 33 levels 1 and 2
- Refurbishment Building 34 ground floor
- Refurbishment Building 36 levels 1 and 2
- New Horticulture Compound
- Refurbishment of Engineering building
- Internal vehicle accessways & parking & associated landscape paths etc
- Associated services enclosures/facilities

UTILITIES

The following utilities are relevant for this project:

- Electricity – Existing HV network is privately owned by CQU
- Water
- Sewer
- Storm water
- Telecommunications
- Fire Authority

2 GENERAL REQUIREMENTS

2.1 GENERAL

2.1.1 DEFINITIONS

Unless the context otherwise requires, the following definitions apply:

- 1 Supply: 'Supply', 'furnish' and similar expressions mean 'supply only'.
 - 2 Provide: 'Provide' and similar expressions mean 'supply, deliver and install' and include all testing, commissioning, tuning, interfaces etc.
 - 3 Proprietary: 'Proprietary' means identifiable by naming manufacturer, supplier, installer, trade name, brand name, and catalogue or reference number.
 - 4 Samples: Includes samples and prototypes.
-

2.1.2 SPECIFICATION OBJECTIVES

The intent of this Specification is to:

- 1 Provide a basis for competitive tendering for suitably qualified tenderers.
 - 2 Allow competent Contractors to price the works.
 - 3 Provide a clear statement of technical and performance requirements against which compliance can be assessed.
 - 4 Provide documentation detailing the scope and quality of the project for the purpose of Client's confirmation of compliance to the Client's project Brief.
 - 5 Define the requirements of the Contractor with respect to Quality Assurance.
 - 6 Define the scope of services including project management, installation engineering, testing commissioning, maintenance, and documentation to be provided.
 - 7 The Specification relies upon the Contractor having the necessary specialist services resources to complete the detailed design, testing and commissioning of the installation in accordance with the design intent and the requirements of the Specification and Design Drawings.
-

2.1.3 DESIGN INTENT

- 1 Intent: The Contract documents are intended to call for complete, operational systems. Provide all items (major and minor), equipment, accessories and incidental work required for the completion of the installation and to ensure full integration.

- 2 Diagrammatic layouts: Layouts of service lines, plant and equipment shown on the drawings are diagrammatic only, except where figured dimensions are provided or calculable. Before commencing work, inspect the site, obtain all necessary measurements and information.

2.1.4 DISCREPANCIES IN DOCUMENTS

Give immediate notice in case of any ambiguity, discrepancy or inconsistency in the Contract document. Should conflict arise between the requirements of Specification, Drawings, mandatory requirements, standards including Standards Australia Codes, then the higher standard or most stringent requirements shall prevail and shall be deemed to have been included in the Contract.

2.1.5 REFERENCED DOCUMENTS/STANDARDS

Use referenced documents/standards which are current editions, with amendments, before the closing date for tenders, except where other editions or amendments are required by Statutory Authorities.

2.2 QUALITY ASSURANCE

- 1 Provide a Quality Assurance plan in order to propose, establish, maintain, monitor and document a quality assurance system covering all aspects of the design, purchase, fabrication, installation and completion of the works. The plan shall be in accordance with ISO 9001/2 (as appropriate).
- 2 Designate a Project Quality Inspector to discharge the quality plan, which must include the following minimum:
 - a Inspect the installation works regularly and provide installation defects reports on regular basis in order to systematically reduce the incidences of installation defects. Submit copy of such defects lists to the Consultant Engineer upon request. This task can be delegated to qualified site foreman if applicable.
 - b Check and sign off all shop drawings (including all third party shop drawings) for conformance to requirements prior to submission. All such documents shall list all non-conformances with the Specification and Drawings.
 - c Check and sign off equipment compliance schedule against the particular Specification requirements and equipment schedules prior to submission. All such documents shall list all non-conformances with the Specification and Drawings.
 - d Check and sign off all samples for conformance to requirements prior to submission. All such documents shall list all non-conformances with the Specification and Drawings.
 - e Check all tests required for proper manufacturing of the equipment.
 - f Check and sign off all manufactured items for compliance prior to dispatch to site.
 - g Check and sign off installation of all items under this Specification. Provide signed off installation ITPs for every elements of installation (minimum floor by floor and/or for every individual area up to 1000sqm whichever is smaller).
 - h Check all materials, welding, joining, terminations, fixing and finishes.

- i** Check and sign off all associated ITPs for all on-site tests required to commission the works.
 - j** Check and sign off operating and maintenance manuals to ensure they contain adequate information to permit systems to be operated by the Client at the end of defects liability period. (Including adequate training and tuition of the nominated Client's representative).
 - k** Check and sign off all associated integrated ITPs for all integrated on-site tests required to commission the works and to be carried out by the various systems.
- 3** All trades works including sub trades works shall form part of overall QA and required to have their ITPs, Shop drawings, technical submissions signed off also by the Project Quality Inspector. This sign off represent all required checking that these works are consistent and compliant with the rest of the works.
- 4** Submit the following documentation:
 - a** Quality System third party certification, if any, to the Standards specified by the Joint Accreditation System of Australia and New Zealand.
 - b** Quality manual detailing, corporate Q.A. policy statement, system element description, register of procedures and project specific ITPs.
- 5** Notice:
 - a** Inspect witness points: If notice for inspection is to be given in respect of parts of the works, advise if and when those parts are to be concealed. Inspect and provide signed off installation ITPs indicating appropriate methods of installation have been carried out.
 - b** Inspection hold points: If notice of inspection is to be given in respect of parts of the works, do not conceal those parts without approval.
 - c** Minimum notice for inspections to be made: 4 hours for inspections full time on-site, otherwise 2 working days for on-site inspections, and 5 working days for local pre-delivery inspections.
 - d** Concealed services: Give notice so that inspection may be made of services to be concealed. Irrespective of Consultant Engineer inspection, inspect the works, sign off relevant ITPs and highlight any noncompliance and/or defects.
- 6** Witness tests: Give sufficient notice so that designated tests may be witnessed.
 - a** Minimum notice for tests to be witnessed is 5 working days for site tests; and 10 working days for local pre-delivery tests.
 - b** Test hold points: Do not carry out designated tests without approval.
- 7** Testing Authorities:
 - a** General: Except for Site Tests, have type tests carried out by authorities accredited by NATA to test in the relevant field, or an approved organisation outside Australia recognised by NATA through a mutual recognition agreement. Co-operate as required with testing authorities.
 - b** Site tests: Use instruments calibrated by authorities accredited by NATA.
- 8** Reports:

- a** General: Submit copies of test reports, including certificates for type tests, showing the observations and results of tests and compliance or non-compliance with requirements.
 - b** Submit Installation Defects Reports carried out on regular basis to the Consultant Engineer upon request.
 - 9** Concealment:
 - a** If tests are to be carried out on parts of the work, do not conceal those parts and do not commence further work on those parts until the tests have been satisfactorily completed and compliance verified.
-

2.3 ELECTROMAGNETIC COMPATIBILITY

- 1** Comply with Australian Communications Authority requirements for electrical and electronics products to limit electromagnetic interference (EMI).
 - 2** Emissions:
 - a** Passive products with C-tick or Regulatory Compliance Mark (RCM) to AS/NZS 4417 (Marking of Electrical Products to Indicate Compliance with Regulations) and the Australian Electrical Equipment Safety Scheme (EESS).
 - 3** Immunity:
 - a** Electrical and electronic apparatus: To AS/NZS 61000.6.1, Electromagnetic compatibility (EMC) – Generic immunity standard – Residential, commercial and light industry, or AS/NZS61000.6.2, Electromagnetic compatibility (EMC) – General standard – Immunity for industrial environments.
 - 4** Harmonics:
 - a** Levels of emissions shall be in accordance with the electricity Distribution Code. Keep the THDI current to 5% Maximum for all individual components/equipment and at least to that level at the first supplying control panel for all components/equipment connected to the switchboard. Provide active harmonic filters within the component, or respective control panel. Do not rely on the external harmonic filters to be installed on the Main Switchboard(s). Refer to the remaining sections for more specific requirements.
-

2.4 BUILDING PENETRATIONS

- 1** Ensure that the penetration of services through walls, floors and roofs is undertaken in a manner consistent with maintaining fire ratings, smoke proof ratings, achieving acoustic performance and achieving weather-proofing, as necessary.

2.4.1 FIRE RESISTING BUILDING ELEMENTS

- 1** Seal penetrations to comply with NCC using a system to AS 4072.1, Components for the protection of openings in fire-resistant separating elements – Services penetrations and control joints.
- 2** Comply with Section 3 of AS/NZS 1668:1, Fire and Smoke control in Multi Compartment Buildings.

- 3 AS4072.1 inter-alia Appendix B and AS1851 inter-alia Section 12 and 13 shall be complied with in as-built documentation to identify all fire and smoke containment compartment walls and barriers. To satisfy specific identification requirements to AS4072.1 Appendix B and AS1851, permanently fixed labelling, tagging and signage of all passive fire and smoke containment systems must be provided for on-site identification. Fire stopping barriers and penetrations shall comply in particular with reference to a service label affixed adjacent to each fire stop barrier and service penetration (or close proximity group) as detailed in Appendix B of AS4072.1 and AS1851. This as-built provision amongst other as-built documentation and manuals is essential as part of handing over process of the project.

2.4.2 NON-FIRE RESISTING BUILDING ELEMENTS

Seal penetrations around conduits and sleeves. Seal around cables within sleeves. If the building element is weatherproof, acoustic rated or subject to pressure, maintain the rating. Acoustically seal penetrations through plantroom walls and floors.

2.4.3 CHASES

Cut, drill and chase as necessary, where possible avoiding finished work. Where chasing of finished work is essential, make good all damage to the original standards of the finishes.

2.4.4 LIMITATIONS

General: Do not penetrate, or chase the following without approval:

- 1 Structural building elements including external walls, core walls, fire walls, floor slabs, beams or columns.
- 2 Acoustic barriers.
- 3 Other building services.
- 4 Membrane elements including damp-proof courses, waterproofing membranes and roof coverings.
- 5 Membranes: If approval is given to penetrate membranes, provide a waterproof seal between the membrane and the penetrating compound.

2.5 REQUIRED SUBMISSIONS

- 1 Provide in good time to allow review without impediment to the programme, including possible amendment and resubmission: -
 - a Documents showing approval of the Authorities whose requirements may apply to the work.
 - b Factory test results where applicable.
 - c All product data, performance test and commissioning results required by this Specification.
 - d Shop drawings for fabrication and installation of all equipment and items supplied.
 - e Manufacturer's product data for equipment, materials, components and systems including:
 - i Technical specifications and drawings.
 - ii Size, arrangement, operating and maintenance clearances.
 - iii Operating weight.

- iv** Type test reports.
 - v** Design verification documentation
 - vi** Performance and rating curves or tables marked with each selection showing for both duty point and for part load conditions the capacity; the energy consumption and power factor; and the sound power level.
 - vii** Motor power requirements (both full load and starting).
 - viii** Recommendations for installation and maintenance.
 - ix** Evidence of compliance with specified product certification schemes.
 - f** Inspection, test and commissioning plans for every section of the works
 - g** Client handover/training proposals.
 - h** Record and installation drawings.
 - i** Operating and Maintenance manuals.
 - j** Maintenance/service records during the defects liability period.
- 2 Give notice before commencing work affected by the submissions, unless the submissions have been reviewed with no exception taken.
 - 3 Do not commence production of equipment, materials, components of equipment, components or systems until submission has been reviewed with no exception taken.
 - 4 Maintain and submit monthly a schedule of shop drawings and submissions to identify all proposed drawings and submissions designated in the following categories:
 - a** Reviewed.
 - b** Submitted, awaiting Review (with submission date and revision number).
 - c** Not yet submitted.
 - 5 Submit electronically (PDF or alternative agreed format).
 - 6 Identify the project, Works Package, supplier/manufacturer, applicable product name, product number, included product options, and relevant specification references.
 - 7 Where selected equipment or systems do not fully comply with this Specification and associated documents, submit details of proposed deviations for acceptance. Where such deviations require any changes to the structure, building works or services, these shall be provided by the Contractor without variation to the Contract. Apart from deviations listed, it will be taken that the materials and equipment proposed meets with all other specified requirements. Review of equipment and materials is undertaken on this basis, and that non-complying equipment/materials may be rejected at any time – even if already installed.
 - 8 Submit amended documents as appropriate, highlighting changes since the previous submission.

2.5.1 SAMPLES

TIMING

- 1 Co-ordinate submissions of samples as listed in the Schedules. Do not cause delays by making late submissions or submitting inadequate samples.

QUANTITY

- 1 Submit a sample of each designated item and 2 copies of supporting documentation. Include ancillary items such as fasteners and mounting brackets. Amend and resubmit samples which are not accepted.

IDENTIFICATION

- 1 Identify the project, Contractor, their agents or supplier, manufacturer, applicable product, model number and options, as appropriate and include pertinent Specification references.
- 2 Include service connection requirements and product certification.
- 3 Identify non-compliances with project requirements, and characteristics that may be detrimental to successful performance of the completed work.

SAMPLE ACCEPTANCE AND RETENTION

- 1 Do not commence work affected by samples until the samples have been accepted. Submit further samples as necessary.
- 2 Keep approved samples in good condition on site, until Practical Completion.
- 3 Incorporate in the works samples, which have been approved for incorporation. Do not incorporate other samples.
- 4 Installed items will match accepted samples throughout the works.

2.6 MATERIALS, EQUIPMENT AND COMPONENTS

2.6.1 GENERAL

- 1 All equipment submissions proposed for use on this project shall be accompanied with a statement from the supplier / manufacturer that it complies with the specified requirements and provide this statement where required by the Consulting Engineer. In the event the equipment / components of equipment do not meet the specified requirement, the relevant change and the specification clauses where the change is proposed shall be highlighted and detailed in the submission. Equipment where no specific changes are highlighted are assumed to meet all specified requirements and if not highlighted may require replacement after installation.

- 2 Provide new materials, equipment and components shall be new, of good quality and fit for purpose, selected for a reasonable service life. Do not provide without approval products that are obsolete, discontinued or about to be discontinued.
 - 3 Identification of a proprietary item does not necessarily imply exclusive preference for the item so identified, but indicates a deemed to comply item. The supplier will still need to confirm the product complies with this Specification requirements
 - 4 Manufacturers: Provide equipment and associated accessories which are the products of established manufacturers regularly engaged in the manufacture of such equipment, who issue comprehensive rating data and certified test data on their products. They also shall have established market record and ability to service the product locally by having local representatives and qualified service personnel
 - 5 Consistency: For the whole quantity of each material or product use the same manufacturer or source and provide consistent type, size, quality and appearance.
 - 6 Safety: Provide all necessary safety devices for the protection of personnel against injury and the protection of plant and equipment against damage including relief valves, belt guards, safety railing, and effective earthing of electrical components, electrical interlocks, warning lights, and signs, alarms and local lighting. Provide lifting eyes as required.
 - 7 Selection: Select, if no selection is given, and transport, deliver, store, handle, protect, finish, adjust, prepare for use, and use manufactured items in accordance with the current written recommendations and instructions of the manufacturer or supplier.
 - 8 Product certification: If products must comply with product certification schemes, use them in accordance with the certification requirements.
 - 9 Provide permanently fixed plates indicating manufacturer, model, serial number, capacity and electrical data for all equipment. Lettering height: 5mm maximum, 1.5mm minimum.
 - 10 Provide materials or products are supplied by the manufacturer in closed or sealed containers
-

2.6.2 FINISHES AND VISUAL ELEMENTS

- 1 Surfaces shall be flat and free of noticeable distortion, warping and twisting.
- 2 Edges and corners accessible to users shall be dressed or rounded (and if a metal object after cutting, filing the metal shall be cryogenically treated) to avoid cutting if touched.
- 3 Pressings shall be accurate in dimension and profile to ensure consistent mating and clearance gaps and pressing shape.
- 4 Mating surfaces between adjacent pressings - flush and accurate to produce smooth distortion free joint.
- 5 Provide protection against damage until Practical Completion.
- 6 Surface shall be clean, undamaged and free of rust and stains

2.6.3 MANUFACTURERS' OR SUPPLIERS' RECOMMENDATIONS

- 1 All equipment and systems shall be installed in accordance with the specific recommendations of the manufacturer, including the provision of ancillary, safety and maintenance devices or facilities deemed necessary. Failure to observe such recommendations may result in equipment or systems being rejected.
- 2 The works require for coordination with the manufacturer to visit the site for the purpose of confirming compliance with the manufacturer's requirements.
- 3 The equipment manufacturer shall be involved in the review of shop drawings to confirm the plant / equipment installation is in accordance with the manufacturer's recommendations both in relation to installation and access for servicing. A letter from the manufacture confirming that the plant / equipment meets the above two requirements shall be provided.

2.6.4 CLEANING AND PROTECTION OF FINISHED WORK

During handling and installation of work at the site, clean and protect work in progress and adjoining work. Apply suitable protective covering on newly installed work where required to ensure protection from damage or deterioration at a time of Completion; otherwise, clean and perform maintenance on newly installed work as frequently as necessary throughout remainder of construction period. Adjust and lubricate operable components to ensure operation without damaging effects.

2.6.5 PROHIBITED MATERIALS

- 1 Do not use any of the following materials:
 - a Lead where the metal or its corrosion products may be directly ingested, inhaled or absorbed, or any lead-based paints or primers.
 - b Chromate paint pigment or chromate water treatment.
 - c Urea formaldehyde foam or materials which may release formaldehyde in quantities which may be hazardous or irritant.
 - d Synthetic mineral fibres except in accordance with Worksafe Australia National Standard and National Code of Practice.
 - e Materials in which chlorofluorocarbons, hydrochlorofluorocarbons or hexa-fluoroacetones have been used as a blowing agent.
 - f Chlorofluorocarbons.
 - g Polychlorinated biphenyls.
 - h Other substances generally known to be deleterious to health or safety or which would adversely affect the works.
 - i Any that would be contrary to achievement of a required project Green Star or other sustainability rating.

2.7 INSTALLATION

2.7.1 GENERAL

- 1 Carry out the work in a proper and workmanlike manner.
- 2 Install equipment and services parallel or perpendicular to building elements. Organise reticulated services neatly. Provide for movement in both structure and service.
- 3 Provide expansion facilities in guide rails, cable trays and supports to accommodate thermal expansion and movement at structural expansion joints.
- 4 Protect equipment from weather and the ingress of dirt, moisture, vandalism and tampering.
- 5 Provide access to all components requiring entry, inspection or maintenance.
- 6 If interruptions to supply of any service are required co-ordinate the shutdowns to the satisfaction of the Superintendent, advise the occupants, minimise inconvenience, and advise when supply is reinstated. Arrange shutdowns out of hours.

2.7.2 NOTICES

- 1 Inspect witness points: If notice for inspection is to be given in respect of parts of the works, advise if and when those parts are to be concealed.
- 2 Concealed services: Give notice so that inspection may be made of services to be concealed. Irrespective of Consultant Engineer inspection, inspect the works, sign off relevant ITPs and highlight any noncompliance and/or defects.

2.7.3 SETTING OUT OPENINGS AND MAKING GOOD

- 1 The Specification Drawings are diagrammatic only and unless otherwise stated shall not be used for determining the precise positions of equipment outlets and like features. The exact location of these shall be determined on site and/or from shop drawings.
- 2 All chases, ducts, recesses and penetrations in structural elements not shown on the building and structural drawings shall be subject to approval.
- 3 Provide the precise location of all openings, fixings and similar items of work required for these works.
- 4 Determine requirements at such times as not to cause delay to the work.
- 5 Check all items to ensure correct positioning.
- 6 Wrongly located or omitted openings, fixings and similar items of work resulting from incorrect or lack of inter trade coordination shall be remedied and paid for under these works.

2.7.4 CONSTRUCTION WORKS PROGRAMME

- 1 Provide a detailed construction programme incorporating activities of every trade (as a minimum for all building services trades). The Works programme shall clearly show the start and completion dates with the typical activities listed below. The list is indicative.
 - a Submission of Project Quality Plan including trades
 - b Submission of shop drawings and major technical submissions.
 - c Shop fabrication
 - d Ordering of major plant and materials.
 - e Delivery of major plant and materials.
 - f Services Rough In dates
 - g Permanent power connection.
 - h Commencement of fitting offs and final terminations.
 - i Authorities acceptance tests (where required).
 - j Commencement of testing.
 - k Commencement and Completion of plant commissioning.

2.7.5 METALWORK

- 1 General: Use metalwork capable of transmitting the loads imposed, and sufficient to ensure the rigidity of the assembly without causing deflection or distortion of finished surfaces. Construct to prevent rattle and nuisance.
- 2 Metal separation: Prevent contact between electrolytically dissimilar metals, by using concealed insertion layers.
- 3 Edges and surfaces: Keep clean, neat and free from burrs and indentations. Remove sharp edges. Provide Cryogenic treatment (cold cutting) of rounded or cut edges after filing to ensure longevity of the metalwork.

2.7.6 FIXING

- 1 Fix all plant to structure in approved manner, either directly or via secondary structure as required. Submit details of types of fixings, locations and loads for approval.
- 2 Fix only light weight items to non-structural building elements.
- 3 Do not pierce waterproof roofs, floors or walls with fixings.
- 4 Fasteners to comply with appropriate Australian / New Zealand Standards.

- 5 Use proprietary corrosion resistant fasteners capable of transmitting the loads imposed, and sufficient to ensure the rigidity of the assembly. Use metal expansion bolts for concrete and masonry. Do not use explosive charge fixings without approval.
 - 6 Corrosion protection: Use corrosion resistant, chemically and electrolytically compatible fasteners. Provide insulating spacers where necessary.
-

2.8 SAFETY IN DESIGN

Safety in Design risk management shall form an integral part of these works and shall conform to the requirements of the Workplace Health and Safety (WHS) Act 2011 & WHS Regulations 2011.

The Safety in Design process shall conform to the State-relevant Compliance and Guidance documentation.

General principles of risk management to be followed shall be as AS/NZS ISO 31000 (2009) Risk Management – Principles and Guidelines.

The National Standard for Construction Work (NOHSC:1016) shall be used to align design risk management practice with construction risk management practice.

The Safety in Design Risk Assessment shall include but not be limited to:

- 1 Understand the range of work activities associated with the intended use of the building / structure as a work place.
- 2 Identify hazards, assess risks (quantify and rank), identify control measures, implement control measures, review and prepare risk register prior to construction of work. Monitor and review throughout construction and operation. The process includes review and acknowledgement of residual risks identified and communicated by the designer as defined by the WHS Act and Regulations.
- 3 Elimination or where this is not reasonable practicable, minimisation of any risk to health and safety of any person. The process includes, recommendation of design alternatives that will eliminate / reduce risks, recommendation of control measures for residual risks; organising participate and facilitating Safety in Design workshops as required.
- 4 Report in writing, on health and safety aspects and risks of the design identified. Implementation of control measures which shall be monitored and reviewed throughout construction and operation.
- 5 Provide at the conclusion of the project all residual risk information, in the form of a Safety in Design File and the related instruction and training as required to ensure an understanding of the safety aspects of the installation.

Where local regulatory practice differs from the above the more onerous shall be used as the basis for compliance.

3 PROJECT REQUIREMENTS AND DESCRIPTION OF WORKS

3.1 GENERAL

Unless varied by Specification Addenda, the requirements set out below in this section are mandatory and no deviation is permitted.

List and allow in your tender form all of the items called in and listed in this section. No deviation from requirements of this section is permitted unless approved by Superintendent in writings during the tender process.

3.2 NOMINATED CONTRACTORS

The approved lift subcontractors for this project are KONE Elevators and Otis Elevators

3.3 ROLES & RESPONSIBILITIES

3.3.1 CONSULTANT ENGINEER

During the construction phase the Consultant Engineer's role is:

- 1 Analysis of schedule of technical data and alternatives offered.
- 2 General review of shop drawings for the purpose of establishing compliance with design intent.
- 3 General review of samples and supportive documentation for the purpose of establishing general compliance with the Specification.
- 4 Review of Contractor's Services trade inspection and test plans (ITPs) for each section of the works.
- 5 Periodic review of the progress of the works on the site, noting any observations with respect to quality of the installation or materials used.
- 6 Issue of any clarification (Consultants Advice) required to interpret the Specification and Drawings.
- 7 Respond to Requests for Information (RFI) via Consultant Advice (CA's) with marked up sketches of drawings if appropriate to assist with interpretation of the Specification or drawings. It is expected that prior to making a request for information the Contractor has studied thoroughly the Contract documents and used its best endeavours to obtain the information.
- 8 Prepare and issue Consultants Advice if required to amend the Specification and or Drawings
- 9 Accompanied by the Contractor's representative, randomly inspect and review test records achieved by witnessing Contractor's Site acceptance test and commissioning demonstrations.

- 10 Review and with attendance by the Contractor's representative, random commissioning data achieved by witnessing Contractor's test demonstrations.
- 11 Review Contractor's Services trade Q.A. records including ITPs.
- 12 Inspect the works at Practical Completion
- 13 Review Client training proposals.
- 14 Review 'As Installed' and operating and maintenance documents.
- 15 Review maintenance records during the defects liability period.
- 16 Inspect the works at the end of defects liability period.

Note: Building Services Engineers 'review' or 'observation' does not relieve the contractor of contractual obligations including ensuring the works are compliant, complete, accurate and correct. 'Reviews' and 'observations' will be carried out only for general conformance with the Design Intent and general compliance with the Contract Documents and are not incorporate detailed review.

3.3.2 OBLIGATIONS OF THE CONTRACTOR

- 1 In writing this Technical Specification the Client expects and relies upon the Contractor to provide its own personnel and its trades possessing specialist trade expertise necessary to complete the works in accordance with this Technical Specification which form part of the overall Contract documentation.
- 2 The Contractor and its subcontractors are to exercise good industry practice and comply with the NCC/BCA and Australian Standards, when carrying out the Works. The Consultant is not liable for any loss, cost, expense or damage arising out of or in connection with the Contractor or its subcontractors not complying with good industry practice, the NCC/BCA and/ or Australian Standards.
- 3 In addition, the Contractor has the following obligations:
 - a To raise in good time, issues requiring design input or clarification from the Consultant Engineer, particularly in respect to:
 - i interpretation of the Specification or drawings;
 - ii problems in complying with the Specification together with suggested alternative/substitutions;
 - iii matters in the Contractor's opinion, are omissions not discovered during the tender process, together with suggested alternatives/substitutions
 - b To ensure that they are in possession of the latest drawings and specifications
 - c To allow the design verification costs of the Consultant Engineer when suggesting alternatives and departures from this Technical Specification and accompanying Drawings.
 - d To certify compliance with Authority requirements.
 - e To pay all fees applicable to the works.
 - f To obtain all Authority permits and certificates to allow the progress of the work. Where notice is required by any statutory Authority having jurisdiction over these Works and/or approval required

there from, give such notice and make application accompanied by such plans and information as may be called for and/or obtain such approvals. Obtain all permits within such time as to ensure no delay to the works and shall pay all fees associated therewith.

- g** To provide manufacturer's and construction drawings of sufficient detail to allow proper fabrication, co-ordination and installation and a scale not less than 1:50; incorporating all post-tender changes to architectural and structural drawings.
- h** To provide samples and prototypes where specified.
- i** To provide manufacturer's factory test records (type tests and witnessed tests) and site installation inspection and sign-off certifying that the completed installation complies with the manufacturer's installation guidelines and requirements
- j** To provide the completed Services installation that is fully and correctly commissioned and fine-tuned and, that is functional and operable efficiently in compliance with all requirements of this Technical Specification.
- k** To design, install and certify the vertical transportation systems in accordance with AS1170.4.
- l** To certify compliance with Contract documents, including all variation instructions, at Practical Completion

3.3.3 CONTRACTOR'S DESIGN RESPONSIBILITIES

Specifically, for specialist services trades, the Contractor shall be responsible for the detailed design activities listed below, as these activities are considered to be normally undertaken through the custom and practice of the industry.

- 1** Comprehensive co-ordination of all services and associated building works forms part of the Works.
Refer Section 3
- 2** Co-ordinate the layout of plant and services with the building layout and structure, and with other plant and services.
- 3** Size and location of penetrations in walls and floors.
- 4** Physical co-ordination of installation with other trades.
- 5** Detailed design (by qualified and registered designers) and installation of seismic restraints
- 6** Provision of electrical loads to electrical trade and compare with design loads cable sizes and circuit protection devices, seeking design direction where discrepancies occur.
- 7** Details of electrical wiring and control diagrams of all equipment supplied showing all interconnections between equipment to enable the necessary wiring to be undertaken.
- 8** Thermal expansion accommodation and anchorage, taking into account final installation details and consistent with specified requirements.
- 9** Mounting, support and fixing details, and fasteners, including any secondary structure which may be required.
- 10** Cable installation details and derating factors.

- 11 Settings for protection equipment, time delays, time switches etc.
 - 12 Capacity, location, design and sizing of cable support, trunking and conduit systems.
 - 13 Acoustic design or modification of actual selected equipment that may require change, to meet with the noise levels specified. Such change to approval of the Consultant Engineer. Specified levels to be achieved with all plant operating.
 - 14 Selection of all anti-vibration mountings to suit the particular application of the mounts
 - 15 Final locations of control sensors, detectors, thermostats, switches, outlets.
 - 16 Services layouts coordinated with architectural, structural, mechanical, hydraulic, audio visual, fire protection, acoustic, fire engineering, etc. Including updating architectural backgrounds and re-designing to suit.,
 - 17 Safety in Design registers and facilitate workshops
 - 18 Spatial co-ordination and advice (allow to liaise with architect & structural engineer to deviate from the plant spatial intent and access intent documented; allow to confirm plant spatial intent shown) and the like.
-

3.3.4 CO-ORDINATION

Comprehensive co-ordination of all services and associated building works forms part of the Works. Ensure that the detailed design undertaken is fully co-ordinated and compatible with the remainder of the project design. These responsibilities shall include:

- 1 Checking of space requirements of equipment and services which are indicated diagrammatically in the Contract documents. Select equipment with dimensions to suit the available space. Prior to ordering equipment, plant, etc. ensure that sufficient space is available, by reference to the Architectural drawings or by site measurement, to accommodate the actual size of the equipment being offered.
- 2 Lay out equipment and services to be accessible for operation, maintenance, and replacement and so as not to interfere with access to other installations.
- 3 Co-ordination of the layout of plant and services with the building layout and structure, and with other plant and services. Co-ordination of the works includes but is not limited to mechanical, hydraulics, structural, acoustic, fire protection and architectural works prior to installation. The contractor shall satisfy themselves that any alterations within the Contract do not adversely affect other work. Neatly group services, with separate areas / layers for crossing of services.
- 4 Comprehensive checking of plant, systems, cabling, etc. capacities and performance against the final arrangement of equipment, e.g. checking capacity of utility connections, checking current carrying capacities / voltage drop, checking for system interfaces and interference in electrical and BAS systems, etc.
- 5 Arrangement and co-ordination of all interconnection testing with other services systems, e.g. fire mode tests, power failure testing, etc.
- 6 Co-ordination of the construction of the installation.
- 7 Physical co-ordination of installation with other trades.
- 8 Interface details with other trades.

- 9 Size and location of penetrations in walls and floors including those fabrics that do not form part of the works under contract but which are necessary to realise the works under contract.
- 10 Co-ordination of in-ground and in-ceiling services routes, including trenching, trays and pipework.
- 11 Management of the preparation and approval of fully co-ordinated drawings including penetration drawings.
- 12 Arrangement and management of inter services / integrated services testing and commissioning.
- 13 Provision of all test results and required certifications to obtain Certificate of Occupancy from the Building Surveyor.
- 14 Co-ordination to ensure that all building and services elements such as air ducts, outlets and light fittings, fire detectors, sprinklers, speakers and other elements are installed in a co-ordinated manner, such that all services can be installed, connected and maintained as indicated on the Architectural Plans.

During the construction period the works require that a competent supervisor is provided on site responsible for the comprehensive co-ordination of all engineering services, including civil, structural, electrical, mechanical, hydraulics, and fire protection services. Responsibility for the proper direction, supervision, control and co-ordination of the work forms part of the works and shall assist all other trades in every way possible in meeting this responsibility.

Should work proceed without co-ordination being undertaken and authorisation by those with such responsibility and should other trades subsequently not be able to effectively proceed with their respective installation, the work initially undertaken shall be removed and reinstalled at the expense of the Contractor.

Any claims for additional costs or delays due to the lack of effective co-ordination by the various trades will be rejected.

3.3.5 ALTERNATIVE PRODUCTS

Should the part of the work be proposed to be done quicker, better or more effectively by substitution of materials or methods other than those specified, the details of such substitutions or alternatives shall be included in the Tender, including a comprehensive Report including any performance and life cycle analysis in support of the proposals.

Where the words 'equal or approved' are used in this Technical Specification, permission may be requested to use a substitute for what is specified, provided it and the relevant manufacturer of the item certify in writing that the substitute is of equal, or better quality, performance and effectiveness to that specified.

Submit for approval full details to allow checking that the alternative products comply with the Contract Documents. Provide a certificate for each alternate product verifying that the proposed alternative product complies with the contract documents. Include all compliance and non-compliance items related to Project Brief, Specifications and Contract Drawings. State if use of proposed alternatives will necessitate alteration to other parts of the works and include consequential costs.

Allow for additional work resulting from the utilisation of an approved alternative product, including additional or revised statutory approvals, changes to adjacent work, re-submission of shop drawings and any costs incurred by the Consultant Engineer in reviewing such alternative products. Allow for Consultant Engineer Review cost based on agreed Market hourly rates. These sums are to be paid irrespective of the alternatives being selected or not.

Proposals for substitution must include details of:

- 1 Changes to adjacent work, if any.
 - 2 Cost differences.
 - 3 Life cycle cost differences
 - 4 Quality differences, specification compliance and advantage to the Client.
 - 5 Lead time and program changes.
 - 6 Certificate of compliance.
-

3.4 EXISTING SERVICES

The Works include contact with all utility companies prior to commencing work when required, to ascertain the precise location of all in ground and above ground services in the vicinity of any proposed excavation or building works. Ensure protection of the services, as necessary, during the progress of the work at all times.

In the case of work in close proximity to in ground services, particularly electrical cables or gas piping, hand digging may be necessary in order to determine the exact location of such services. Electronic tracing of in ground services may be required. Any damage to existing services shall be made good as part of the work of the Contract, at no additional cost to the Client.

3.5 STAGING OR SEQUENCING OF WORKS

Make due allowances for staging and sequencing, including temporary works, staged commissioning, out-of-hours works, etc.

Assess the impact of the staging or work sequence on the services involved, notwithstanding that the services documents may not identify such staging or sequencing.

3.6 DESCRIPTION OF WORKS

Whether or not particular works are described in the Technical Specification all items and materials needed for the complete works are required and shall be installed unless clearly excluded.

The project includes the detailed design, supply, delivery, installation, commissioning, tuning, testing, placing into service, maintenance, warranty and defects liability of materials, labour and plant of the services systems in accordance with this Technical Specification. All components and systems shall be complete in every respect and tested and commissioned unless otherwise noted specifically.

The work shall include all necessary minor and incidental work required to implement the intent and meaning of this Technical Specification and associated drawings. Whether or not the words 'supply and install' appear in this Specification or on the Drawing, unless clearly excluded, all items of equipment for the complete installation are required and shall be supplied and installed.

Provide advance notice of any proposal to diverge from previous detailing and/or specifications stating reasons for proposed change. All work shall be carried out in a manner to the approval of the consultant.

3.6.1 GENERAL

- 1 Allow to decommission, make safe and remove any existing redundant services to be deleted within the area of works.
- 2 Site visit and familiarization with the site conditions and scope of work required under this contract and making required allowances to complete the works.
- 3 All necessary negotiations including formal submissions as required with all Authorities/utilities having jurisdiction including obtaining of design approval and final certification of the installed systems by an independent certifier approved by the Authority. Payment of all fees associated with the authorities/utilities connections and permits for the installation.
- 4 Provision of detailed safety in design analysis and ensure the installation is in compliance with the Site's safety in design proposal including all regulatory requirements.
- 5 Allow for all requirements in relation to alternatives proposed for the project. Allow for Consultant Engineer review cost based on agreed market hourly rates. These sums are to be paid irrespective of the alternatives being selected or not.
- 6 Management and coordination of overall services shop drawing process including providing timely input into the process to allow fully coordinated shop drawing production for all services. Active engagement and provision of nominated staff for input into the multidisciplinary coordination workshops in relation to shop drawing production as well as onsite installation methodology and integrated commissioning of multi-disciplinary interfaces.
- 7 Coordination with the Electrical trade on final mechanical equipment selections and power requirements of mechanical services switchboard to ensure correct submain cabling from the MSB to mechanical services switchboards. Provide all lugs, cable glands and gland plates for final connection and termination by Electrical trade to the mechanical services switchboards. Any submains/sub-circuit to secondary switchboards/control panels shall be included in the respective services Works Packages.
- 8 Provision of QA plan including inspection, testing, commissioning ITPs for all systems installed.
- 9 Preparation of "For Construction" and "Shop Drawings" coordinated with other services prior to commencement of works including services layouts at the same scale as the design drawings.
- 10 Detailed design and preparation and submission of design schematics and calculations prior to shop drawing.
- 11 Preparation and submission of shop drawings and equipment schedules prior to construction.
- 12 Provision of all samples and technical submissions.
- 13 Identification and coordination of penetrations, provision of fire/smoke sleeves and sealing of all wall, floor, ceiling penetrations to the required fire/smoke/acoustic rating.
- 14 Provision of vibration isolation for all rotating equipment.
- 15 Provision of seismic restraints, including design.
- 16 Provision of all painting and finishes including identification and labelling schedules,

- 17 Provision of As-Built drawings and O&M Manuals in compliance with a uniform overall Site Services Documentation System.
 - 18 Preparation and supply of complete and detailed Operating and Maintenance Manuals which include all commissioning data in both hard and soft copy format. Ensure Operation and Maintenance Manuals clearly indicate on an individual equipment basis a schedule of maintenance routines, inspections, preventative maintenance procedures, test surveys and records conducted on the fire and smoke protections systems installed as outlined in AS1851. Generic type procedures / inspections are not acceptable.
 - 19 Provision of all training for staff in the operation and preventative maintenance of the systems. Training of the nominated building facility management staff to the level of understanding of day by day operation of the services systems.
 - 20 Twelve (12) months statutory testing, servicing, maintenance, defect liability and warranty of all services systems/equipment installed under this contract from the date of Practical Completion.
-

3.6.2 INTER SERVICES WORKS

- 1 Provision of wire mesh screens on the inside face of the lift well to suit ventilation opening at the top of lift well.
 - 2 Provision of lighting and power outlets within the lift wells
 - 3 Provision of Access Control and CCTV cabling from lift cars to connection box(es) supplied
 - 4 Installation only of Access Control card reader in lift COP' where required
 - 5 Install and connect CCTV cameras in lift car
 - 6 Provision of Public Address and Hearing Augmentation cabling from lift cars to connection box(es).
 - 7 Install and connect Public Address and Hearing Augmentation interface in lift car.
-

3.6.3 OVERALL SYSTEMS

- 1 Provision of a complete Lift services system inclusive of all equipment, controls, certifications, finishes, landing stations, landing indicators, car operating panels/ displays/buttons, interfaces as further detailed herein and/or shown on the drawings
- 2 Note that the lift supplier is to allow for their standard packaged lift that best fits the available lift shaft dimensions.

3.6.4 SCHEDULE OF LIFTS

DESIGNATION		Building 33	Building 36	NTB
No. of units in Group		One (1)	One (1)	One (1)
Lift Numbering		Lift No. 1	Lift No. 1	Lift No. 1
Code Compliance		AS1735.1.1, 1735.1.2/EN81-20, AS1735.12 & 1735.19	AS1735.1.1, 1735.1.2/EN81-20, AS1735.12 & 1735.19	AS1735.1.1, 1735.1.2/EN81-20, AS1735.12 & 1735.19
General	Load (kg)	2,250	1,600	2,250
	Load Rating	Goods Passenger	Goods Passenger	Goods Passenger
	Passengers (No.)	30	21	30
	Speed (m/s)	1.0	1.0	1.0
	Machine type	G'Less	G'Less	G'Less
	Control System	Conventional	Conventional	Conventional
	Drive System	VVVF	VVVF	VVVF
	Front opg's	G, 1 & 2	G & 1	G, 1 & 2
	Rear opg's	Nil	Nil	Nil
	No. Stops	3	2	3
	Stretcher Carrying Capacity	Yes	Yes	Yes
	Emergency Lift Requirement	Nil	Nil	Nil
Communication Interfaces	Access Control Interface	Yes	Yes	Yes
	BMS Interface	Yes (Low Level)	Yes (Low Level)	Yes (Low Level)
	Remote Monitoring Interface	Yes, refer to section 8	Yes, refer to section 8	Yes, refer to section 8
	CCTV Interface	Yes, CAT 6a and GPO on lift car top	Yes, CAT 6a and GPO on lift car top	Yes, CAT 6a and GPO on lift car top
Car sizes	Inside Car Width (mm)	1,800	1,600	1,900
	Inside Car Depth (mm)	2,500	2,100	2,400
	Inside height to false ceiling (mm)	2,600	2,600	2,600
	Roof pocket	Nil	Nil	Nil

DESIGNATION		Building 33	Building 36	NTB
Doors	Type	2-PC	2-PC	2-PC
	Clear width (mm)	1,300	1,100	1,300
	Clear height (mm)	2,400	2,100	2,100
	No. entrances per lift car	One	One	One
	Sill Load Rating (EN81-20 5.7.2.3.6)	Goods Passenger (Fs =0.6 x g _n x Q)	Goods Passenger (Fs =0.6 x g _n x Q)	Goods Passenger (Fs =0.85 x g _n x Q)
	Emergency Access Doors	Nil	Nil	Nil
Liftwell	Clear Width (mm)	2,810	2,500	2,950
		Refer to Architectural drawings	Refer to Architectural drawings	Refer to Architectural drawings
	Clear Depth (mm)	3,000	2,400	2,925
	Pit depth (mm)	1,800	1,600	1,800
	Travel (mm)	Refer to Architectural drawings	Refer to Architectural drawings	Refer to Architectural drawings
	Headroom (mm)	4,800	4,600	4,800
	Pit to solid earth	Yes	Yes	Yes
	Pit Sump Water Sensor	Yes	Yes	Yes
	Lighting	Yes, with emergency fitting top and bottom	Yes, with emergency fitting top and bottom	Yes, with emergency fitting top and bottom
Machinery Space	Separate from liftwell or MRL	MRL, controller cabinet located on uppermost served floor.	MRL, controller cabinet located on uppermost served floor.	MRL, controller cabinet located on uppermost served floor.
Power	Automatic Rescue Device	Yes, refer to section 7.1.4	Yes, refer to section 7.1.4	Yes, refer to section 7.1.4
	Operation on Standby Generator	No	No	No
	Regenerative Drive	Yes	Yes	Yes
	Minimum Energy Efficiency Certification (ISO 25745-2)	As per NCC Section J Requirements	As per NCC Section J Requirements	As per NCC Section J Requirements

DESIGNATION	Building 33	Building 36	NTB
Operations	— Exclusive Service — Fire Service Recall Facility	— Exclusive Service — Fire Service Recall Facility	— Exclusive Service — Fire Service Recall Facility — Hazardous Goods operation as specified elsewhere

3.6.5 LANDING STATIONS

APPLICATION	LIFTS WITH CONVENTIONAL CONTROL
Quantity	Single Lifts: One riser
Design	Landing button station shall be of the Standard VT Works Package flush mount design, incorporating buttons as detailed in this specification. If there is not enough room to mount the landing button station on the liftwell wall due to the available space being within 500mm of a corner; the VT contractor shall allow for extended cabling so that the landing button station can be mounted on the adjacent wall greater than 500mm away from the corner. The architectural drawings shall be reviewed so that the necessary allowances are made.

3.6.6 LANDING INDICATORS

APPLICATION	LIFTS WITH CONVENTIONAL CONTROL
Design	Hall lantern shall be of the flush mount design. Standard VT Works Package design
Display	LCD Direction Indicator and LCD Car Position Indicator.

3.6.7 CAR OPERATING PANEL

APPLICATION	LIFTS WITH CONVENTIONAL CONTROL
No. panels per car	Two (2) one Main and one Auxiliary
Design	Standard VT Works Package Design to meet the requirements of AS1735.12 and incorporate buttons as detailed in this specification. Where this arrangement is provided physical buttons as per the requirements of this specification shall be provided for the door open, door close and alarm button.
Door Hold	Via push button
GPO's	Provide one GPO in each lifts Main COP. The finish of the GPO faceplate shall match that of the COP.

3.6.8 LIFT CAR DISPLAYS

APPLICATION

LIFTS WITH CONVENTIONAL CONTROL

Type	Standard LCD digital position and direction car information display.
Number and Location	One on each COP, located between 1,800 and 2,100 above the car floor level.

3.6.9 BUTTONS

Type	Round shaped dual illuminating with Standard English Braille located on the button. Button shall incorporate a 15mm high tactile legend with Braille and shall incorporate an illuminated halo which encircles the pressel. The face of the button / pressel shall be of a finished stainless steel finish. Buttons shall be vandal resistant to EN81-71 category 2. Buttons shall meet AS1735.12 requirements.
Colours	Buttons shall be backlit in white and when pressed illuminate blue or other approved colour combinations. The contrast of the buttons shall comply with AS1735.12 The emergency communication button shall illuminate yellow when activated as per the requirements of AS1735.19.

3.6.10 FIXINGS

Type	Secret fixings or screw fixings – Delco tamper proof or approved equivalent
Finish	Any visible fixings that cannot be secret shall be the same finish as the faceplate

3.6.11 KEY SWITCHES

Type	Key switches shall be of the high security type specifically designed for the lift industry. The core of the key switch shall be removable which allows for easy recombination. Switch Mounts for fire service shall be red in colour. Switch Mounts for all other keys shall be silver in colour. Engraving shall be provided on the Switch Mount
On Landing Stations	Provide the following Key-Switches: — Fire service at Ground Level landing station. — NTB lift only - Hazardous Goods Operation Key-Switch to allow the lift when on Hazardous Goods Operation, to be “sent” to its selected floor (3 position spring return – off, on & send)

On Main COP	Provide the following Key-Switches: — Exclusive service (3 position – off, on, and park). — Fire service (2 position – off and on). — NTB lift only - Hazardous Goods control for the lift (2 position – off & on).
Fire Service Keys	A set of fire service keys shall be labelled and hung in a secured and locked closure located adjacent to the Fire Indicator Panel (FIP). The enclosure shall incorporate a lock that is keyed to a standard fire brigade 003 key.

3.6.12 DOORS

Type	Horizontal sliding
Door Sills	Anodised aluminium Stainless steel for NTB lift
Entrance Monitoring	3D lightray with colour indication. Lightray shall flash red upon door closing and green upon door opening. Lightrays shall meet the requirements of AS1735.12
Landing Doors	Landing doors shall be provided with access devices at all landings. Access devices shall be in compliance with EN81-20 requirements.

3.6.13 LANDING ENTRANCES

Frame Finish	No. 4 Linished Stainless Steel
Frame Type	Full depth including projection beyond wall finish.

3.6.14 PROTECTIVE BLANKETS

Type	Padded fire resistant
Quantity	One set for each lift

3.6.15 CAR GUIDE SHOES

Type	Roller guides or silent running sliding shoes.
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3.6.16 TRAVELING CABLES

Allowance	In addition to those required for normal lift function allow: 1 6 x shielded twisted pair cables, 4 x Cat 6a cable shall be provided for the following: a Emergency Phone b Access Control c CCTV d EWIS/PA – if required
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Spare	Allow 10% spare single core, twisted pairs and double insulated power cables.
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3.6.17 LIFT CAR INTERIORS & APPOINTMENTS

LIFT CAR FINISHES

A weight allowance of 500kg is to be allowed for lift car finishes (including floor finish).

ENTRANCE WALL	No. 4 Linished Stainless Steel
WALLS SIDE	To be selected from manufactures standard range of finishes
WALLS REAR	To be selected from manufactures standard range of finishes
WALLS FRONT RETURN	To be selected from manufactures standard range of finishes
SKIRTINGS	To be selected from manufactures standard range of finishes
HANDRAILS	To be selected from manufactures standard range of finishes and comply with AS1735.12
BUMP RAILS	To be selected from manufactures standard range of finishes
FLOOR	To be selected from manufactures standard range of finishes
CEILING/FALSE CEILING	To be selected from manufactures standard range of finishes
LIGHTING	High efficiency LED lamps. Lights shall be automatically controlled as detailed in the controls section of this specification.
PASSIVE CAR WALL VENTILATION	As required by AS1735

APPOINTMENTS AND DOOR FINISHES

LANDING STATION FACEPLATES	<i>No. 4 Linished Stainless Steel</i>
Landing Indicator faceplate	No. 4 Linished Stainless Steel
Landing Indicator jewel (if applicable)	White polycarbonate

Car Operating Panel(s)	No. 4 Linished Stainless Steel
Car Doors	No. 4 Linished Stainless Steel
Landing Doors	No. 4 Linished Stainless Steel

3.7 SHOP DRAWING AND CONSTRUCTION DRAWING REQUIREMENTS

3.7.1 GENERAL

The term shop drawings includes manufacturing and For Construction drawings and shall mean complete drawings showing all details of installation and fixing of the items concerned, and shall include any necessary explanatory notes or specifications.

Prepare, and submit for review before commencing manufacture or installation, Construction Shop Drawings from which the subject works shall be built. Shop drawings shall be dimensioned, showing layout and installation of all plant and equipment, including relationship to building structure and other services. No factory or site work, other than manufacture of standard equipment or controls, shall proceed until such drawings have been reviewed and marked as satisfactory. Show all dimensions and note which are based on site measurement. Confirm dimensions on site prior to preparation of drawings. Allow for adjustments and fitting on site. Indicate whether site dimensions have been verified. Confirm site dimensions prior to fabrication. No extra charge or compensation will be considered for differences between actual dimensions and dimensions indicated on Contract Documents. Do not delay job progress.

Where plant and equipment is to be located within or above the structure, submit sketch plans at an early stage of the project to demonstrate that the spatial allocations are sufficient for effective installation, testing, commissioning and lifetime maintenance or replacement.

Co-ordinated penetration drawings shall be submitted in sufficient time for review. Any failure to submit requirements within sufficient time for revision of structural drawings, resulting in re-working of the structure or re-ordering of structural components shall be responsibility of Contractor and shall not constitute any claim for variation or delay. All penetration measures shall relate to grid lines, co-ordinates or relative levels.

Prepare and submit drawings of penetrations and “built-in” components in the same form and manner as described for the processing of shop drawings. All penetration measures shall relate to grid lines, co-ordinates or relative levels.

3.7.2 SHOP DRAWING MINIMUM REQUIREMENTS:

Provide drawings generally in accordance with the design drawings format (CAD /Revit) supplied. Each drawing shall be submitted on a Contractors Title Block. Provide the following information on the title block:

- 1 Project identification/Contractor's and Subcontractor's Names/ Date of preparation of submission, and all revisions, if applicable
- 2 Shop drawing number and title of item to which the shop drawing refers
- 3 The name(s) of the contractors staff or approved representatives responsible for preparing the shop drawings
- 4 Statement or stamp of approval by the Contractor, which shall signify that he has seen and examined the shop drawing as conforming to the Specification.

Shop drawings shall be submitted in standard metric sheet sizes and included as noted on the following table. All shop drawings shall be dimensioned and submitted for review. Provide the following:

- 1 General
 - a Design of connections not indicated on the Contract Documents
 - b Details of all interfaces with other services and utilities
 - c Location and identifier for all equipment.
 - d Co-ordination with other trades services (lighting, sprinkler pipework, cable trays, speakers etc.) for spatial co-ordination with building structure and other services with details for restricted locations
 - e Label and signage schedules
 - f Maintain current sets of drawings on site and progressively record variations which lead to creation of as-built drawings.
 - g Access to and removal of all plant including locations and sizes of access doors to be installed in finished building surfaces.
 - h Plant room layout plans 1:50 scale, with sections and details at 1:20 scale. All services Floor plans at 1:100 scale, with detailed sections.
 - i Major equipment details
 - j Explanatory notes and specifications.
 - k Site survey references, grid locations and set out.
 - l Methods of assembly, including junctions and trim to adjoining surfaces, fittings and accessories and methods of installation, including fixings, anchorages and sealants.
 - m Legend of Symbols
- 2 Sections
 - a 1: 50 scale.
- 3 Structural Coordination
 - a Layout and details of services cast in concrete (for co-ordination and review by structural engineer / architect).
 - b Structural support details and methods of fixing to structure.

- c** Construction loadings (for structural engineer review).
- d** Lifting point loads (for structural engineer review)
- e** Details of required penetrations in structural elements.
- f** Details of items to be cast-in, built-in or fixed by others.
- g** Provision for movement of the work of the base structure.
- h** Foundations, plinths, chases, ducts, pits and penetrations through structure.

4 Controls & Electrical

- a** Complete control wiring diagrams showing all electrical controls, relays, cut-outs, timing devices, inter-locks, fuses, over-loads, contactors, solenoids, starters, etc. with all items clearly identified as to type and function, and 'as installed' switchboard layout drawings.
- b** Switchboard and control panel layouts and schematics with terminal numbering.
- c** Control logic diagrams with settings.

5 Schematics

- a** piping, tray and conduits, electrical and controls.

Submit computations with shop drawings to demonstrate the adequacy of the submitted design/detail. Computations shall be in a form acceptable to the Consulting Engineer and Authorities having jurisdiction over the works.

Allow a minimum of 5 working days for the shop drawing review after delivery to the consulting engineer for appraisal.

If amendments are required from the Consulting Engineers review, shop drawings are to be revised and resubmitted. This process shall continue until the shop drawings are deemed satisfactory. Review of shop drawings by the Consulting Engineer refers to general design intent only and does not relieve the Contractor of responsibility for contract requirements, or dimensional accuracy of the information presented in the shop drawings.

Please note when shop drawings review comments states “satisfactory subject to comments”, resubmit the revised drawings with comments incorporated within a period of no longer than 2 weeks. However this should not stop the installation works.

Once a drawing is accepted as satisfactory, the contractor shall maintain a current, set of the final shop drawing on site for review at all times. The Contractor shall progressively record all changes to equipment and services layouts, wiring and any other items during the construction period, which may have been incorporated into these works. These drawings shall be used as the basis for the creation of as-built drawings.

3.7.3 LIFT SHOP DRAWINGS

1 The following lift Shop Drawings shall be provided as a minimum:

- a** Lift layouts.

- b** Lift car interiors.
- c** Lift landing entrances.
- d** Lift car and landing faceplate appointments.
- e** Lift notices, labels and signs.
- f** Any other item of equipment visible to a normal user of the finally installed equipment.

2 These Shop Drawings shall show the following information:

- a** Lift layouts:
 - i** The layout shall show the following minimum information to demonstrate that the Works will meet the specified requirements and to ensure that the end user is provided with sufficient detail for future technical reference in the form of “as installed” data and that the design team members have the information needed to design their elements associated with the lift; some information may be provided in an alternative format to the Layout if done so in an acceptable manner.
 - ii** Such information as may be required by the architect and structural engineer to ensure that the lift enclosure and structural supports will be dimensionally and structurally adequate for the proposed lift, including where applicable building grid references and relationships thereto and North orientation and enclosure details and penetration details and monorail and lifting eye and beam and hatch and plinth and vision window details and the like and loads imposed on the building including appropriate impact allowances and earthquake loadings and indication of which loads may act simultaneously and in which direction. Elevations on the entrance enclosure wall shall be viewed from the lobby and not from the liftwell.
 - iii** Such information as may be required by the Consultant Engineer to ensure that the HVAC requirements of the lift are provided.
 - iv** Such information as may be required by the Consultant Engineer to ensure that the power supply requirements of the lift are provided.
 - v** Such information as may be required by the Consultant Engineer to ensure that the fire prevention and detection requirements of the lift are provided.
 - vi** Such information as may be required by the Consultant Engineer to ensure that the hydraulics requirements of the lift are provided.
 - vii** Such information as may be required by the Consultant Engineer to ensure that where applicable the access control and communication and surveillance requirements of the lift are provided.
 - viii** Code classification such as Passenger or Class of Goods.
 - ix** Major equipment details including where applicable rated load and speed and acceleration, machine type, motor size and rating, control system type and model, safety gear type, Power Door Operator type and model, guide rail sizes, rail loading data for Goods lift, hydraulic ram size, etc.

- x** Liftwell plan including clearances and dimensional relationships between adjacent elements and showing finished dimensions of lift car and entrance and liftwell enclosure and major steel work and general landing entrance frame profiles and pit door location.
 - xi** Where a separate machine room then the machine room plan showing all lift equipment Including machine and controller and governor and circuit breaker and location and size of major associated plant such as machine room air conditioning equipment.
 - xii** Vertical section through the liftwell and machine room if separate showing major vertical dimensions and relationships Including where applicable entrance height, inside car height, car false ceiling height, pit depth, false floor height, pit door height, levels served, level designations, reduced levels of finished floor levels of floors served, vertical dimensions between all levels served, total travel between terminal floors served, liftwell headroom between top level served and underside of structure over liftwell, height of machine room structure and monorails and lifting eyes therein, and height of major associated plant such as machine room air conditioning equipment.
 - xiii** An elevation on each different entrance wall viewed from the lobby showing finished landing entrances and landing appointments and their relationships to finished floor levels and to each other.
- b** Lift car interiors:
- i** The drawing shall show the visible finishes and dimensions of all visible elements of the car and car doors Including the floor, walls, skirtings, handrail, false ceiling and ceiling.
 - ii** Finish specifications that completely describe the material.
 - iii** Car faceplate appointments as dimensioned shapes with their relationship to the car floor and walls and description as to their function e.g. main car operating panel, auxiliary operating panel, etc.
 - iv** Fittings and descriptions such as light fittings and lamp specifications, and surveillance camera if any.
 - v** Ventilation details including natural and mechanical.
 - vi** How mechanical ventilation avoids directly drawing air from adjacent natural ventilation slots at top of car walls.
 - vii** Complying regarding fire hazard properties.
 - viii** Construction behind Visible panels on those walls not containing entrance and handrails allowing removal of panels and handrails from within lift for repair and replacement.
 - ix** Constructed so as to allow Complying continued use of lift until panels and handrails replaced (excepting only handrail requirements for disabled use which may apply).
- c** Lift landing entrances:
- i** The drawing shall show an elevation from the landing and a section through the architrave and a sectional plan through the entrance.

- ii The profile in plan and elevation of the architraves shall be shown and dimensioned including jamb depth, projection beyond finished wall, face width, return beyond finished wall, method of fixing and relationship of head to jambs where visible.
 - iii Radii and projections shall be dimensioned.
 - iv Construction whether welded or bolted shall be specified.
 - v Landing doors shall be shown.
 - vi Material and finish shall be specified.
 - vii The elevation shall show the adjacent landing faceplate appointments as shapes with a description as to their function e.g. call button faceplate, hall lantern faceplate, etc, and shall dimension the relationship of the faceplates to the floor and adjacent landing entrance.
 - d Lift car and landing faceplate appointments:
 - i The drawings shall show in large scale the faceplates and all appointments and fixings including third party access control fixtures.
 - ii Material and finish shall be specified.
 - iii Surface engraving method and infill shall be detailed.
 - iv Method of fixing and hinging shall be detailed.
 - v Projecting elements shall be dimensionally defined.
 - vi Font style shall have been previously ascertained from the architect.
 - e Lift notices and labels and signs:
 - i The drawing shall show in large scale all visible labels and signs.
 - ii Material and finish shall be specified.
 - iii Method of fixing shall be detailed.
 - iv Surface engraving method and infill shall be detailed.
 - v Font style shall have been previously ascertained from the architect.
 - f Other lift items of visible equipment.
 - 3 The drawings shall show all other items of equipment visible to the user that are not specified above, and items that the building manager or designate may use such as monitors and displays and controls.
-

3.8 SITE PROVISIONS & BUILDING WORKS IN CONNECTION

This clause is provided to highlight the project requirements in relation to the Site provisions as well as Building Works in Connection BWIC and to ensure that they have been allowed for part of the Contractor Tender submission. Contractor is responsible for demarcation of tasks and responsibilities between various trades in relation to these items to suit his/her delivery methodology.

- 1 Site Provision include but not limited to;
 - a Hoisting and Lowering;
 - b Scaffolding;
 - c Temporary Services – Water, Electricity;
 - d Site facilities including Sheds;
 - e Power & Fuel for Testing and Commissioning.
-

3.8.1 *BUILDING WORKS IN CONNECTION*

GENERAL

Allow for the following;

- 1 Demolition:
 - a Services decommissioning, disconnection from supply, capping and sealing.
 - b Services demolition and removal.
- 2 Ground Works:
 - a Services trenches; excavation and backfilling.
- 3 Concrete:
 - a Chasing, coring, cutting and making good.
 - b Placing, casting in and protection of pipe sleeves and conduits.
 - c Water-proof curbs round floor penetration in plant rooms.
- 4 Masonry:
 - a Chasing, coring, cutting and making good.
- 5 Structural Steel:
 - a Lifting beams.
 - b Platforms and walkways for maintenance to equipment.
- 6 External Walls
 - a Metal louvre grilles complete with bird mesh for air intakes and exhausts.
- 7 Internal walls
 - a Setout of cutouts including cutout and trimming of openings
- 8 Doors:
 - a Access doors and hatches for maintenance of Services equipment.
- 9 Ceilings:

- a Co-ordination of penetrations.

10 Dry wall/ Plenum Construction

- a Co-ordination of penetrations.

VT SPECIFIC

The following items are not intended to be a complete description of the building works in connection, however, are to be noted as being provided to the VT Contractor to allow for a complete and safe installation. The Contractor is responsible to coordinate all required building works with the builder.

- b Liftwell enclosure constructed to the tolerances required by the selected VT Contractor.
- c Liftwell enclosure constructed to the requirements of the VT Contractor's shop drawings including but not limited to pits, headrooms and entrance openings.
- d Penetrations through walls and slabs, and wiring sleeve as required.
- e Installing finished building surfaces up to installed door frame equipment, once equipment fixed by the VT Contractor.
- f Supply and Install protection of the landing entrance openings prior to the VT Contractor commencing on site.
- g Installation of overhead beam or lifting eye at the top of the liftwell
- h Building in of the VT Contractors cast in items such as bearing plates and inserts flush with inside face of the liftwell wall.
- i Suitable structure in lift entrance pathway for fixing landing sill supports.
- j Well flushing other than in path of car entrance.
- k Pit sump including cover
- l Liftwell vermin proof ventilation grill

3.9 SERVICES INTERFACE WORKS

Carry out the works outlined under all contract documents including this Specification to achieve a fully operational installation including all required interfaces between various systems. Whilst the responsibilities for interface works and division of works is solely with Contractor the following information is provided for guidance only. Contractor has full control to alter and modify the trade allocation/demarcation of interface works to suit the Contractor's overall programme and delivery methodology in order to achieve a fully integrated, interfaced and operational whole services installations, systems and sub-systems.

3.9.1 TRADE PACKAGE INTERFACES

Table below indicate key areas where different work packages will interface with each other. Column 1 indicate the disciplines/work packages identified and column 2 & 3 outline the key interface works between these two work packages.

	VT Responsibilities	Others
VT 1 / Mech 2	— Confirmation of all heat loads that will arise in lift machine areas for determination of ventilation louvre sizing. — Supply and installation of wire mesh screens on the inside face of the lift well to suit ventilation opening at the top of lift well. — Provide, install and connect voltage free contacts to the BMS termination point (4 points per equipment).	— Provision of vermin proof and weather proof ventilation grille and fire damper on the outside of the lift well to suit the ventilation opening at the top of lift well. — Supply and Install connection box outside the top floor controllers of each lift and to each LMR, for the BMS and connection thereto, and cabling from connection box to the buildings BMS infrastructure. — Supply and Install connection box at suitable location at top landing of each escalator, for the BMS and connection thereto, and cabling from connection box
VT 1 / Elec 2	— Coordinate the final location of Lift control panels switchboards for submains reticulation. — Termination of submains cables for non MRL lifts. Final connection of submains for MRL lifts part of VT Works Package. — Install and connect CCTV cameras in lift car — Access Control and CCTV cabling from lift cars to connection box(es) supplied — Provision for the housing of an access control card reader in lift COP's, and landing stations where required. — Provision of a dual SIM GSM unit to operate the emergency phone. SIM cards to be included for in the VT package for the duration of the DLP period.	— Supply 3-phase sub-main(s) from building main switchboard to lift electrical termination points in accordance with requirements of AS3013. Confirm the type and size of cables with the VT Trade Contractor. — Coordinate the final location of Lift control panels switchboards for submain reticulation. — Final connection of submains for MRL lifts part of VT Works Package — Supply of CCTV camera and any enclosure and mounting bracket — Co-ordinate the wiring / cabling system requirement for access control and CCTV — Supply and Install connection box outside the top of each lift shaft and cabling from connection box to the buildings access control and CCTV infrastructure. — Supply, install and commission access control card reader. — Provision of interface from security system to lift lift controller for access control and alarm monitoring. — Provision of lighting including emergency lighting above the MRL lift controller panel typically installed at the top floor served by the lift. — Provision of lift management system cabling from building managers / security control room to top of shaft for MRL lifts.

	VT Responsibilities	Others
VT 1 / Fire 2	— Provision for the lift to recall to a landing and turn off all power in response to a sprinkler activation signal from the FIP. AS1735.1.2 (EN81-20) – if required	— Provision of fire detection system in lift wells above the controller. — Provision for sprinkler protection in the lift pit.
VT 1 / Hyd 2	— Detail location of pit sumps on shop drawings — Install moisture sensor in dry sump and connect cabling within lift pit to connection box(es)	— Supply moisture sensor for installation by VT inside the lift pit. — Provision for water sensor alarm / BMS cabling from connection box(es) and wiring thereto.

3.10 DESIGN AND REGISTRATION OF PLANT

- 1 Comply with the occupational health and safety authority's requirements applicable to the Works in respect of:
 - a Plant registration of the Works; and
 - b Plant design registration
- 2 Design registration shall be obtained prior to the commencement of the Works on site. Obtain Plant registration prior to the Works being commissioned for use (for a Contractors lift; or Principal/Client at completion). Pay all fees related to design registration and plant registration.
- 3 The following information shall be provided in a timely manner to demonstrate that authority requirements are being met:
 - a Plant registration data required for the registration of each complete item of plant shall be provided prior to use or completion of the Works; and
 - b Plant design notification certificate.
- 4 In circumstances where plant registration is required by the authority to be made by the Principal/Client, prepare the application on their behalf and assist in its submission. Follow up on the completion of the registration application and obtain a copy of the final registration certificate to include in the O&M Manual.
- 5 Provide a "Safe to Operate" certificate prior to the lift being made available for use by a Contractor or an end user.

3.11 USE OF LIFT FOR CONSTRUCTION

- 1 For the purposes of this clause "Construction Use" shall mean the carrying of people or materials in or on a lift other than for the purposes of installing and testing and commissioning by the VT Works Package and "legal" use shall mean the satisfaction with the condition of the lift by the Authority having jurisdiction over the acceptable use of a lift for the use to which it is put.
- 2 Should a lift be used for Construction Use prior to Practical Completion of that part of the Works the following shall apply;

- a** The Consultant shall have no responsibility in regard to any aspect of such use.
- b** The lift shall be used safely and legally including separation and guarding from any adjacent lift and supervision of its use and adequate maintenance and emergency communication and the rescue of trapped passengers.
- c** The lift shall be adequately protected against damage, wear and tear for the use to which it is put.
- d** The lift and its environment shall be restored to the same condition and cleanliness as applied before such use commenced.

4 DESIGN AND PERFORMANCE CRITERIA

4.1 GENERAL

The following standards are applicable throughout the project.

- 1 National Construction Code (2022) including all relevant Australian Standards called in the NCC
- 2 Relevant Occupation/Workplace Health and Safety legislation
- 3 The Environmental Protection Authority
- 4 WorkCover
- 5 Any other Authority having jurisdiction over all or part of the installation to ensure that the equipment and installation, when manufactured and installed, will comply with the rules and regulations.

4.2 AUSTRALIAN/NEW ZEALAND STANDARDS AND CODES

The following standards are applicable throughout the project as a minimum. Where items of equipment are required the relevant Australian Standard to the equipment will apply. If clarification required, the Contractor shall seek clarification at the time of tender.

Advise all subsequent amendments to applicable standards and codes as they come into force during the work and seek direction regarding compliance for this project.

Where Australian Standards and Codes do not exist, the appropriate British or NFPA Standard or Code shall apply. Where this Specification expressly requires Standard higher than or different from those applicable under the relevant Standards and Codes requirements, this Specification shall be followed.

The latest version of the standards will be used including all applicable amendments.

Refer the requirement of NCC for the nominated Australian Standards.

STANDARD/CODE	TITLE
AS 1428.1	Design for access and mobility – General requirements for access – New building work
AS 1428.2	Design for access and mobility – Enhanced and additional requirements – Buildings and facilities
AS 1428.3	Design for access and mobility – Requirements for children and adolescents with physical disabilities
AS 1735.1.1	Lifts, escalators and moving walks – General requirements

<i>STANDARD/CODE</i>	<i>TITLE</i>
AS 1735.1.2	Lifts, escalators and moving walks - Lifts for the transport of persons and goods –Passenger and Goods Passenger lifts
AS 1735.1.3	Lifts, escalators and moving walks - Safety rules for the construction and installation of lifts - Examinations and tests - Design rules, calculations, examinations and tests of lift components
AS 1735.11	Lifts, escalators and moving walks – Fire-rated landing doors
AS 1735.12	Lifts, escalators and moving walks - Facilities for persons with disabilities
AS 1735.19	Lifts, escalators and moving walks – Remote Alarm on Passenger and Goods Passenger Lifts
AS/NZS 3000	Electrical installations (known as the Australian/New Zealand Wiring Rules)
AS/NZS 3439.1	Low-voltage switchgear and control gear assemblies - Type-tested and partially type-tested assemblies
AS/NZS 4431	Guidelines for safe working on new lift installations in new constructions
CIBSE	Commissioning Codes
EN81	Safety rules for the construction and installation of lifts
ISO 14798	Lifts (elevators), escalators and moving walks — Risk assessment and reduction methodology
ISO 18738	Lifts (Elevators) – Measurement of Lift Ride Quality
ISO 8100-20	Lifts for the transport of persons and goods – Part 20: Global essential safety requirements (GESRs)
NCC	Section E & J

4.3 SEISMIC RESTRAINTS

4.3.1 DESIGN STANDARD

- 1 NCC - Building Code of Australia.
- 2 AS1170.4 - Structural design actions - Earthquake actions in Australia

4.3.2 DESIGN PARAMETERS

The following design parameters shall be used to design the Engineering Services equipment and fastenings as per AS1170.4:

Importance Level	Refer to structural engineers documentation
Site Sub-Soil Class	Refer to structural engineers documentation

Design and install the scope of works to withstand the earthquake forces determined from the above design criteria in accordance with the requirements of the BCA and AS1170.4.

4.4 DESIGN AND PERFORMANCE CRITERIA (VT SERVICES)

4.4.1 PERFORMANCE

DEFINITION AND MEASUREMENT

Dwell	Interval from doors fully open to doors start to close.
Car Ride Quality	The criteria to be quantified for car ride quality, and their symbols are as follows: Horizontal acceleration of the car: — front to rear - A_{hf} — side to side - A_{hs} Vertical acceleration and deceleration of the car - A_v Vertical “jerk”, meaning the time rate of change of acceleration of the car - J Peak to peak, meaning the sum of the absolute values of the maximum positive and negative values over the time duration specified.

UNITS AND QUALIFICATIONS

SYMBOL	UNITS	QUALIFICATIONS & COMMENTS
A_{hf} and A_{hs}	m-g	1 m-g = $g/1,000 = 0.00982 \text{ m/s}^2$ Measured in the range 1 to 10 Hz. Filtering – not ISO. Measured at a sampling frequency of not less than 100 Hz. Measured peak to peak over whole journey for full travel, where adjacent peaks are less than 1 second apart. Measured in both directions of car travel. Measured with load anything less than 350 kg.
A_v	m/s ²	Applicable over journey while vertical velocity not constant. Measured in both directions of car travel. Measured with both rated load and with 350 kg load.
J	m/s ³	Measured over journey while vertical acceleration not constant. Measured in both directions of car travel. Measured with both rated load and with 350 kg load.

PERFORMANCE REQUIREMENTS

REQUIREMENT	LIFTS
Minimum - Vertical acceleration (mps ²)	0.4
Maximum - Vertical acceleration (mps ²)	0.9
Av : tolerance on acceleration and deceleration rates (%)	-0, +5
J : maximum jerk rate (mps ³)	1.6
Ahf and Ahs : maximum horizontal acceleration inside lift car measured as specified (m-g)	20
Rated hoist motor starts per hour, measured with doors set at dwell necessary to achieve specified number of starts (No.)	180
Rated power door operator motor starts per hour when driving full door system (No.)	480
Dwell after answering landing call or combined car and landing call: adjusted as instructed at each floor on an individual lift basis, adjusted throughout Defects Liability period to Specification requirements: Minimum (secs) Maximum (secs)	2.0 10.0
Dwell after answering car call (secs) - adjusted throughout Defects Liability period to Specification requirements	2.0
Dwell after re-establishment of door protection device or operation of door close button in car operating panel (secs) - adjusted throughout Defects Liability period to Specification requirements	0.5
Maximum door open time	3
Maximum door close time	4
Floor levelling accuracy +/- (mm)	5 to 10
Minimum visual and audible landing indication prior to arrival of lift car and doors starting to open (secs).	3.5

4.4.2 ADDITIONAL REQUIREMENTS FOR ACCESS BY THE DISABLED

Lift designated as passenger lifts shall comply with AS1735 Part 12

- Where there is a difference between a minimum requirement of AS1735 Part 12 and a requirement of the Specification that also complies with AS1735 Part 12 the requirement of the Specification shall be complied with.
- Where there are options within AS1735 Part 12 they shall be selected as follows:

4.2 4.2.b – beams.

4.3 4.3.a – at least 3 seconds of advance warning of arrival at the floor.

8.5	8.5.b.(i) – non-oral sounds.
8.6	8.6.1 – locate on the same wall as the Car Operating Panel
10.3	10.3.(b) – internally illuminating control buttons.

4.4.3 ELECTRICAL INFORMATION

- 1 Power Factor Correction shall not be less than 0.95.
- 2 Provision of fire rated submains for all lifts
- 3 Contractor to coordinate lift maximum demand and maximum cable size with Electrical Contractor

4.4.4 NOISE AND VIBRATION REQUIREMENTS

- 1 Measures necessary to maintain noise and vibration requirements shall be implemented Including use where appropriate of isolating pads or mountings, loaded to manufacturer's recommendations, Including hoisting machine, controller, penetrations between machine space and liftwell if appropriate, switchgear, sheave, guide shoes, door mechanism, rope hitch and chain.
- 2 Prior to Practical Completion and as part of the normal testing and commissioning procedures the lift machinery noise shall be subjected to acoustic measurements in order to determine their compliance with the Specification. Sound level measurements shall be made outside normal working hours.
- 3 Sound levels shall be measured in "A" weighted decibels or dB(A).
- 4 The Equipment or Equipment machinery noise shall be described by the maximum sound level occurring during the measurement period or L_{max} value. The sound level measured shall be representative of the overall sound pressure level for the frequency range bounded by and including the 31.5 Hertz and 8k Hertz octave bands.
- 5 Where the sound level varies considerably with time, as is the case with lift pass-by noise and other single event levels, the L_{max} sound level shall be measured on 5 to 10 occasions and arithmetically averaged.
- 6 In occupied areas such as lift lobbies and inside lift cars, measurements shall be made at positions where people would normally be located, approximately 1.0 to 1.5 metres from any wall or flat surface.

REQUIRED NOISE AND VIBRATION MEASURES – LIFTS

- 1 The following table describes some required measures and the maximum allowable sound level at various locations within the building under different operating conditions, subject to a tolerance of 2dB(A).

APPLICATION	L_{MAX} DB(A)
Inside lift car measured at least 1 metre away from any flat surface with ventilation switched on while the ambient noise level in lift lobby is at least 5dB(A) below the event limit.	55
Lift levelling into floor with doors opening and with doors closing	58

<i>APPLICATION</i>	<i>L_{MAX} DB(A)</i>
Lift accelerating and decelerating	55
Lift running at contract speed	55
Inside lift foyer, lift passing floor while the ambient noise level in lift lobby is at least 5dB(A) below the event limit	55
Inside machinery space with machine running while lift performs journey of at least one floor run	82
Inside liftwell while lift performs journey in each direction between bottom and top terminal floors	70
Vibration emanating from Equipment shall not be apparent in occupied areas of the building other than plant rooms.	

GUIDE RAIL SUPPORTS

To minimise structure borne noise from the lift equipment, guide rail brackets and trimmer beams shall be positioned vertically as follows:

- 1 Coincident with floor slabs at each level.
- 2 Midway between floor slabs.

4.4.5 ENERGY EFFICIENCY

Lift shall comply with NCC 2022 Section J7D8

4.5 ENVIRONMENTAL RATING SYSTEMS REQUIREMENTS

Refer to ESD Specification for details

5 PAINTING, PROTECTION AND IDENTIFICATION

5.1 GENERAL REQUIREMENTS

- 1 Select painting and finishes in accordance with the area environmental conditions.
- 2 Review the risk assessment and statutory and authority requirements completed as part of the system testing, commissioning and handover requirements and provide labelling and signage as determined.
- 3 Where exposed to view paint exposed equipment, cable trays, ductwork and pipework
- 4 All black steel surfaces and surfaces subject to corrosion shall be primed with at least two coats of primer and protected.
- 5 Protect all items of work during dispatch and whilst on site during progress of installation.
- 6 Fabricated pipe and steelwork shall be delivered to site prime coated
- 7 If exposed to view, paint new services and equipment including in plant rooms, except chromium, anodised aluminium, GRP, UPVC, stainless steel, non-metallic flexible materials and normally lubricated machined surfaces. Repaint proprietary items only if damaged.
- 8 Do not combine paints from different manufacturers in a paint system.
- 9 Remove or protect fixtures, equipment, surfaces and labels before starting to paint, and re-instate after completion of painting.
- 10 Repair factory finishes if damaged with identical finishes.
- 11 Finish visible joints made by welding, brazing or soldering using methods appropriate to the class of work (including grinding or buffing) before further treatment such as filling, painting, galvanising or electroplating.
- 12 If galvanised surfaces have been cut or welded after galvanising, prime the affected area using zinc rich organic binder AS/NZS 2312.
- 13 Before applying coatings to metalwork, complete cutting, drilling and other fabrication, remove all grease, oil and other contaminants and prepare surfaces to AS 1627.0
- 14 Use relevant Australian Standards for the following finishes; Galvanising, Electroplating, Anodising, Thermoset powder coating, Two-pack liquid coating, Air-drying enamel, Stoving enamel

5.2 CORROSION PROTECTION

5.2.1 GENERAL

- 1 Apply corrosion protection measures to all metallic surfaces.
- 2 Select an appropriate protective coating system to suit atmospheric corrosivity category as defined in referenced Standards/Codes.
- 3 Fastenings are to be protected to have a corrosion resistance equivalent to the parts fastened.
- 4 All welding is to be completed before any corrosion protection treatment is applied.

ATMOSPHERIC CORROSIVITY CATEGORY	TYPICAL EXTERIOR ENVIRONMENTS	TYPICAL INTERIOR ENVIRONMENT
C1: Very Low	Few alpine regions	Heated or air conditioned buildings, such as most of commercial buildings. Unheated or non-air conditioned buildings in semi-sheltered locations remote from marine or industrial influence
C2: Low	Dry, rural areas as well as other regions remote from the coast or sources of pollution. Typical areas occur in arid and rural inland regions, most inland cities and suburbs of cities on sheltered bays.	Unheated or non-air conditioned buildings, where some condensation may occur, such as warehouses and sports halls, can be in this category. Proximity to the coast is an important factor.
C3: Medium	Coastal areas with low salinity	Food processing plants, laundries, breweries, printing works, dairies
C4: High	Sea-shore (calm)	Swimming pools, dye works, paper manufacturers, foundries, smelters and chemical processing plants, livestock buildings
C5-I: Very High (Industrial)	Sea-shore /offshore	-
C5-M: Very High (Marine)	Shoreline (severe surf)	Adjacent to acidic processes
T: Inland Tropical	Non-coastal tropics	-

5.2.2 PROTECTION FOR STEEL SURFACES

- 1 Apply corrosion protection to meet the performance criteria scheduled in the table below.
- 2 Standards of surface preparation procedures and of corrosion treatments are to meet the specified criteria and the referenced Standards/Codes for the relevant application.
- 3 Surface preparation procedures and coating systems for corrosion protection are to meet all relevant requirements of the referenced Standards/Codes.

ATMOSPHERIC CORROSIVITY CATEGORY	SURFACE PREPARATION	COATING SYSTEM
C1: Very Low	Class Sa 2½ Blast Cleaning 'Very thorough' (commonly called 'near white') blast cleaning which removes millscale, rust and foreign particles to the extent that only traces remain.	1 st Coat – zinc phosphate alkyd primer, 75 microns thick 2 nd Coat – full gloss alkyd finish, 50 microns thick 3 rd Coat – full gloss alkyd finish, 50 microns thick Or Factory applied zinc primer and thermoset powder coated finish
C2: Low	Class Sa 2½ Blast Cleaning	1st Coat – zinc phosphate alkyd primer, 75 microns thick 2nd Coat – full gloss alkyd finish, 50 microns thick 3rd Coat – full gloss alkyd finish, 50 microns thick Or Factory applied zinc primer and thermoset powder coated finish
C3: Medium	Class Sa 2½ Blast Cleaning	1st Coat – Organic zinc primer, 75 microns thick 2nd Coat – polyurethane, 100 microns thick Or Factory applied zinc primer and thermoset powder coated finish
C4: High	Class Sa 2½ Blast Cleaning	1st Coat – Organic zinc primer, 75 microns thick 2nd Coat – polyurethane, 100 microns thick Or Factory applied zinc primer and thermoset powder coated finish
C5-I Very High Industrial	Manufacturer's recommendation	To approval
C5-M Very High Marine	Class Sa 2½ Blast Cleaning	1st Coat – Organic zinc primer, 75 microns thick 2nd Coat – polyurethane, 100 microns thick Or Factory applied zinc primer and thermoset powder coated finish
T Inland Tropical	Class Sa 2½ Blast Cleaning	One coat inorganic zinc coating, 75 microns thick Or zinc or aluminium thermal spray metallic coating.

5.2.3 PROTECTION FOR NON-FERROUS METALLIC SURFACES

- Standards of surface preparation procedures and of corrosion treatments are to meet the specified criteria and the referenced Standards/Codes for the relevant application.

- 2 Surface preparation procedures and coating systems for corrosion protection are to meet all relevant requirements of the referenced Standards/Codes.

5.3 PAINTING

5.3.1 LOW VOC

The following additional requirements are to be met if required by the Green Star Schedule in Schedules.

- 1 All painted surfaces to use low-VOC paints.
- 2 VOCs are to be in accordance with The Australian Environmental Labelling Association, Inc. Standard No: AELA 23-2005 'Australian Voluntary Environmental Labelling Standard Architectural and Protective Coatings'. Conformance with VOC levels listed in this standard (refer to table below) shall be demonstrated by providing test reports from laboratories accredited to carry out the relevant tests and/or calculations and appropriate documentation of production methods and quality controls.

<i>PRODUCT TYPE</i>	<i>MAXIMUM VOC CONTENT (G/LITRE)</i>
Latex Primer for galvanised iron and zincalume	60
Exterior latex undercoat	55
Interior latex undercoat	65
Interior gloss	75
Interior semi-gloss	16
Exterior gloss	75

- 3 VOC limits on architectural coatings covered. Allowable levels include water content in the formulation.
- 4 For solvent-based coatings the paint shall not contain VOCs in excess of 200g/litre. For recycled paints, the VOC level (averaged across batches) in the paint must not exceed 100g/litre.
- 5 The VOC content of the paint shall either be calculated from the VOC data for each of the raw materials or, experimentally by ASTM D3960, as qualified the Australian Environmental Labelling Association, Inc. Standard No: AELA 23-2005 'Australian Voluntary Environmental Labelling Standard Architectural and Protective Coatings'.
- 6 Where the raw material is a mixture of compounds, some of which contain VOCs, the VOC content of the mixture may in turn be calculated from the VOC content of the individual components. Where this is not known, it must be determined by the methodology detailed in AELA 23-2005.

5.3.2 PREPARATION OF SURFACES

- 1 Preparation: all surfaces shall be prepared by thorough cleaning of all dirt, grease, scale, welding flux and corrosion in accordance with AS/NZS 2312.

- 2 For galvanised surfaces that have been subsequently cut or welded, prime the affected area with organic zinc rich coating for protection of steel to AS/NZS2312, two packs or accepted alternative.
 - 3 Prime/undercoat all surfaces. Galvanised and non-ferrous surfaces must be etch primed.
-

5.3.3 APPLICATION

- 1 First coat: apply the first coat immediately after the substrate preparation and before contamination of the substrate can occur.
 - 2 Apply all paint systems in strict accordance with manufacturer's recommendations.
 - 3 Finish: each coat of paint or finish shall be uniform in colour, gloss, thickness and texture and free of runs, sags or blisters.
-

5.3.4 FINISHING

- 1 Concrete Bases: 2 coats synthetic emulsion paint
 - 2 Equipment Panels/Switchboards: 2 final coats full gloss enamel
 - 3 Equipment: Factory spray application
 - 4 Other surfaces: 2 final coats full gloss enamel
-

5.4 COLOURS

Paint colours nominated are to be in accordance with AS 2700 and AS 1345.

5.5 IDENTIFICATION AND LABELLING

5.5.1 GENERAL

- 1 Mark all systems and equipment including wiring, piping, control panels, switchboards, ~~ductwork~~, equipment cabinets, controls, and fittings to that they are readily identifiable.
- 2 Equipment requiring special maintenance procedures or presenting occupational health and safety hazards: provide permanent labels complying with statutory requirements.
- 3 Locate labels so that they are easily seen and either attached to, below or next to item referred.
- 4 Labelling to match installation documentation schedules.
- 5 For systems containing hazardous materials provide labelling in accordance with AS1318.
- 6 Affix hazardous equipment with warning signs.
- 7 Labels shall be permanent.
- 8 Type: select from the following:

- a** Stencil with black or white lettering contrasting with background.
 - b** Engraved two-colour laminated plastic mechanically fixed.
- 9** Adhesive labels and laminated plastic labels shall not be used where exposed to the environment.
- 10** Painted labels shall be in the same finish paint type as the installation.

6 INSPECTION, TESTING, COMMISSIONING, HANDOVER & DEFECTS LIABILITY PERIOD

6.1 GENERAL

This section sets out the engineering and quality assurance requirements for the completion of the installation, inspection regime, testing, commissioning, handover and defects liability period.

6.1.1 DESIGN

- 1 Complete the detail engineering of the testing commissioning and handover of the works to the specified requirements.
 - 2 Comply with applicable authority design guidelines and codes of practice and requirements.
 - 3 As a guide, commissioning guidelines as detailed by CIBSE Commissioning Codes shall be used as a minimum to undertake the commissioning of services listed in this section.
-

6.1.2 STANDARDS & GUIDELINES

Refer to Design and Performance Criteria Section

6.1.3 TEST & COMMISSIONING TYPES DEFINITION

Except where otherwise defined, the following apply:

- 1 Test: Generic term used to define any testing activity
- 2 Functional Test: A site test aimed to verify performance of an equipment, system or integrated systems against the design criteria, acceptance criteria and/or manufacturer's performance requirements
- 3 Prototype tests: Tests carried out on an item or assembly for development or to demonstrate suitability for the project.
- 4 Type tests: Tests carried out on an item identical with a production item before delivery to the site and witnessed by an approved independent testing authority.
- 5 Production tests: Tests carried out on manufactured equipment before delivery to the site.
- 6 Factory Tests: Tests carried out at the Contractor's or its sub-contractor's premises prior to Factory Acceptance Testing and delivery to site.
- 7 Factory Acceptance Tests: Tests carried out at the Contractor's or its sub-contractor's premises and witnessed by the Consulting Engineer prior to delivery to site.
- 8 **Site completion tests & commissioning:** Functional Tests carried out before the date for practical completion on installations and/or systems which have been completed and commissioned, to demonstrate that the installations and systems, including components, controls and equipment operate

correctly, safely and efficiently and meet performance and other requirements and are integrated with connecting systems.

- 9 **Site acceptance tests and commissioning:** Functional Tests to be witnessed by the Principal's Representative/Consulting Engineer carried out before the date for Practical Completion on installations and systems after they have been already successfully completed and commissioned via Site Completion tests and commissioning in order to demonstrate that the installations or systems, including components, controls and equipment operate correctly, safely and efficiently and meet performance and other requirements and are integrated with connecting systems.
- 10 **Deferred tests:** Acceptance tests carried out during the defects liability period (e.g. performance tests to suit seasonal climatic conditions and/or occupancy)

6.1.4 TESTING & COMMISSIONING INSTRUMENTS

- 1 General: Except for site tests, have type tests carried out by authorities accredited by NATA to test in the relevant field, or an approved organisation outside Australia recognised by NATA through a mutual recognition agreement. Co-operate as required with testing authorities.
- 2 Site tests: Use instruments calibrated by authorities accredited by NATA.
- 3 All instrumentation used in the commissioning of the installation shall be managed in accordance with the requirements of AS 3912 or NEBB.
- 4 Reference instruments are to be NATA certified.
- 5 All commissioning results to include instrument calibration documentation

6.1.5 DOCUMENTS REQUIRED FOR OCCUPANCY AND COMPLETION

Prior to occupancy and completion of the vertical transportation services the following items (documents and certificates) shall be provided by the VT Works Package as follows: items in the column "Certificate of Occupancy" (required by the Building Surveyor) are only required to the extent required by the relevant State and Commonwealth laws.

ITEM DESCRIPTION	CERTIFICATE OF OCCUPANCY	CONTRACTUAL COMPLETION	NOTES
Certification that the lift installation meets the requirements of the NCC (BCA) Including clauses C2.10, C3.10, Section E3, and any fire engineering requirements for the project	Req'd	Req'd	Include test certificates for wall and floor coverings in lift cars where applicable
Certification that any lift installation landing entrances have been provided and installed in accordance with AS1735 parts 1, 11 and 12.	Req'd	Req'd	
Plant design notification and registration documents as required by the Occupational Health and Safety Act and Plant Safety Regulations	Req'd	Req'd	

Hazard and Risk documents as required by the Occupational Health and Safety Act and Plant Safety Regulations	Req'd	Req'd	
Copies of all inspection, commissioning and test documents setting out details of all tests conducted and the results and data resulting there from.	Req'd	Req'd	Including specified performance tests.
Certification of the entrance fire rating properties.	Req'd	Req'd	
Certificate of Electrical Safety and compliance with AS3000	Req'd	Req'd	
Certification that the installation complies with all legislative requirements and relevant published Standards Including AS1735		Req'd	
Certification that the installation complies with the contract documents Including this Specification		Req'd	
Completed and Approved Installation manual as specified Including as-built drawings in the manual and electrical circuit drawings on site		Req'd	

6.2 PRACTICAL COMPLETION REQUIREMENTS

- 1 Notwithstanding any other provisions of the Contract with regard to the granting of Practical Completion, the following requirements, listed in order of importance / priority, shall be completed and completion submissions made prior the date of Practical Completion.
- 2 Minimum submissions required for application for Occupancy Permit
- 3 Submission of signed off testing and commissioning schedules to confirm that Testing and Commissioning has been finalised and that all plants and systems operate in a safe, stable and automatic manner under all conditions of full and partial load - full commissioning data.
- 4 Submission of Authority approvals including Statutory Authority approvals and Certification of Compliance with the NCC
- 5 Submission of Certification that works comply with the Contract Documents
- 6 Operating Instruction and Maintenance manual sections covering all Safety Services as required by regulations
- 7 Submission of correctly and completely executed ITPs including test results completed in accordance with the Contractor Quality Plan and the satisfaction of the Consultant Engineer
- 8 Adequate training and instruction of the Principal's representatives in safe operation of the plant
- 9 Official Equipment manufacturer's acceptance certification for the installation
- 10 Completed Operating and Maintenance manuals and 'as built' drawings

11 The Defects Liability period may be extended in the event:

- a** Specific systems fail to achieve the desired outcomes after two attempts are made to rectify the drawback. These specific systems include those that are fire and life safety or business / mission critical to the Principal.
- b** If there have been more than three (3) attempts to rectify the defect, in which instance a further 12 months of monitoring will be required after the defect is rectified to monitor that the desired outcomes are achieved without further shortfalls.

6.3 MANAGEMENT OF INSPECTION, TESTING, COMMISSIONING, HANDOVER AND DEFECTS LIABILITY PERIOD COMMITMENTS

6.3.1 GENERAL

1 Manage the whole process so as to:

- a** Ensure complete execution of Project Specific Quality Plan and specifically Installation and Commissioning ITPs are completed progressively from commencement of the project.
- b** Test and verify the operation of individual items of plant and equipment, sub-systems, systems and the overall installation including the interfaces with other building services systems forming part of the project.
- c** Create and record accurately and verify all “As Built “drawings progressively prior to services being concealed by construction of ceilings, wall cavities and shafts.
- d** Ensure completed Operating and Maintenance manuals inclusive of comprehensive system descriptions, operating modes and associated control functionality, tailored and project specific manufacturer’s literature.
- e** Responsive defects rectification process during the construction phase as well as post Practical Completion during Defects Liability Period.
- f** Ensure Responsive approach towards maintenance and operational fine tuning during the Defects Liability Period

6.3.2 SEQUENCE OF ACTIVITIES

- 1** Submission of Project specific Project Quality Plan including generic and project specific ITPs within the first quarter of the project programme after site possession.
- 2** Ongoing inspection of installation works by relevant Quality Manager and/or his nominated staff for meeting project quality requirements including preparing Inspection reports (listing all observed installation defects) not longer than fortnightly and also checking that all Installation ITPs are signed off.

- 3 Initiate Safety in Design process and documentation.
- 4 Evidence of adequate commissioning programme allowance in construction programme.
- 5 Evidence of currency of calibration of equipment to be used in commissioning procedures.
- 6 Installation and commissioning Inspection and Test Plans ITPs in soft copy format i.e., excel and PDF. As well as 1 hard copy to be supplied on site on day of commissioning.
- 7 Amendment to ITP as necessary and re-submission at least 1-2 weeks after the original submission.
- 8 Submission of controls manufacturer's functional descriptions elaborating on the methodology proposed to implement the requirements of the control strategies. Manufacturer's controls functional descriptions shall be prepared and submitted within the first quarter of the project programme after site possession with any re-submission required within 3 days.
- 9 Nomination of those to be involved in commissioning including individuals with responsibility for management, engineering and field testing and commissioning.
- 10 Liaison with and briefing of Authorities to ensure that the connection to their assets and commissioning procedures and resultant data provided for their approval is consistent with their requirements and records.
- 11 Preparation of periodic (minimum monthly) progress reports on testing and commissioning.
- 12 Early and active engagement of Independent Commissioning Agent ICA where applicable..
- 13 Submission of Operating and Maintenance Manual (for Safety Services and, for preliminary draft of the overall manual) within the first quarter of the project programme after site possession with any re-submission required within 3 days. Confirmation of acceptance of proposed Electronic Operating and Maintenance Manual software/platform.
- 14 Submission of detailed commissioning procedures for review by the Consultant Engineer within the first quarter of the project programme after site possession with any re-submission required within 3 days.
- 15 Rectification of all defects likely to interfere with testing and commissioning.
- 16 Manufacturers' certificates for all proprietary items, to confirm that the installation complies with their installation and maintenance requirements prior to start-up of equipment. Certificates shall also be provided at commencement of equipment installation to confirm their requirements with respect to installation are being provided for.
- 17 Manufacturers' test certificates of all capacity/performance verification checks carried out at factory prior to dispatch to site.
- 18 Commencement of "As Built" drawing production
- 19 Carry out preliminary site completion testing and commissioning until achievement of correct operation and performance. Providing statement that the commissioning activities for each system are completed.
- 20 Site completion testing and commissioning of major plant and equipment, complete systems, interfaces between systems and finally all systems which interact together under normal or emergency conditions.
- 21 Site completion Integrated testing and commissioning of integrated multi systems (including multi-disciplinary systems) and their associated interfaces.

- 22 Site acceptance testing & commissioning witnessed by the Consulting Engineer of the selected systems and Integrated systems
 - 23 Rectification of any further defects and deficiencies found during testing and commissioning.
 - 24 Provide notification of completion of remedial work in sufficient time to permit re-inspection as necessary before the intended date for re-testing.
 - 25 Repetition of sequence of activities as necessary until works comply with acceptance criteria as set out in the ITPs.
 - 26 Final handover of Final Operating and Maintenance Manuals.
 - 27 Certification that works comply with the Contract Documents and with the NCC and Statutory Authorities and are ready for granting of Practical Completion.
 - 28 Accurate and verified (by QA manager)"As Built" drawings.
 - 29 Complete Operating and Maintenance Manuals (Electronic system/package and, unless instructed otherwise, hard copies)
 - 30 Warranties.
-

6.3.3 SUBMISSIONS

Required submissions include but not limited to the following;

- 1 Installation and commissioning Inspection and Test Plans ITPs.
- 2 Evidence of periodic (weekly) Quality Inspection/defects reports
- 3 Testing and commissioning procedures including integrated commissioning procedures.
- 4 Evidence of Safety In Design process during the construction phase is completed where applicable
- 5 Evidence of currency of calibration of equipment to be used in commissioning procedures.
- 6 Records of all pre-commissioning checks and final commissioning data. All records/data to be certified as being conducted in accordance with agreed and approved procedures.
- 7 Manufacturers' certificates for all proprietary items, to confirm that the installation complies with their installation and maintenance requirements prior to start-up of equipment. Certificates shall also be provided at commencement of equipment installation to confirm their requirements with respect to installation are being provided for.
- 8 Manufacturers' test certificates of all capacity/performance verification checks carried out at factory prior to dispatch to site.
- 9 Accurate and verified (by QA manager) "As Built" drawings
- 10 Complete Operating and Maintenance Manuals (Electronic system/package and, unless instructed otherwise, hard copies)
- 11 Warranties

- 12 Maintenance Plan and Schedules following acceptance of technical submission and installation minimum 4-8 weeks prior to Practical Completion with priority with respect to timely completion given to plan and schedules covering Safety Services.

6.3.4 INSPECTION AND TEST PLANS

- 1 Inspection and test plans are to be prepared specifically for the project but may be based on or customised from generic ITPs. All commissioning result forms shall be completed, including the following:

Tested By: Please print Witnessed By:

Signed: Signed:

Date: Date:

Test Device used:

Serial Number:

Date of Calibration:

Method of Testing:

- 2 List acceptance criteria for each element and sub-element of the installation in ITPs. Acceptance criteria to include:
- a Specification details in regard to materials, construction methods, physical requirements, performance and operational requirements.
 - b Schedule of Technical Data in respect of make, model and performance details of equipment.
 - c Shop drawings.
 - d Control logic and diagrams.
 - e Authority inspection and testing requirements.
 - f References to specific clauses of this Specification for acceptance criteria

6.3.5 COMMISSIONING PERSONNEL

- 1 Provide only qualified registered / licensed personnel specific to the trades as required by local regulations, and where regulatory requirements do not apply, personnel certified by appropriate industry bodies.
- 2 Submit a schedule of key commissioning personnel together with their qualifications prior to undertaking commissioning activities on site.

6.4 TESTING & COMMISSIONING

6.4.1 *PRE-ENERGISATION TESTS*

Prior to energisation of any system with electrical connections, conduct the following tests:

- 1 Tightness of screwed and bolted connections.
- 2 Physical integrity.
- 3 Correct phasing. Check phase rotation is consistent throughout the project and is the same as the supply.
- 4 Motor rotation.
- 5 Insulation resistance tests.
- 6 Test operation of all trips, interlocks, motor driven devices, contactors and control circuits and devices by instigating or simulating inputs.

6.4.2 *TESTS*

- 1 Specified consecutive hoist and door motor starts with lift fully loaded cycling between adjacent floors with door dwell set to zero seconds.
- 2 Safety gear test.
- 3 Speed test.
- 4 Static balance test.
- 5 Levelling and re-levelling tests.
- 6 Overload test.
- 7 Control system performance tests.
- 8 Brake tests,
- 9 Single brake shoe testing,
- 10 Light ray testing,
- 11 Fire service testing,
- 12 Emergency phone testing,
- 13 Performance Guarantee tests.
- 14 VT Works Package standard field test data.
- 15 Ride quality tests.
- 16 Acoustics and vibration tests.

- 17 Inspections to ascertain compliance with the Specification as may be amended throughout the contract.
- 18 Tests as detailed in EN81-50 & EN81-58 (applicable to lift installations) - BS EN 81-50 Safety Rules For The Construction And Installation Of Lifts - Examinations And Tests - Part 50: Design Rules, Calculations, Examinations And Tests Of Lift Components; BS EN 81-58 Safety Rules For The Construction And Installation Of Lifts - Examination And Tests - Part 58: Landing Doors Fire Resistance Test

Documentation of inspections and tests conducted

ARCHITECT'S INSPECTIONS

Inspection of Equipment and finishes for correctness, appearance and quality of finishes. Documentation of inspections and tests conducted.

6.4.3 COMMISSIONING

Operate the installation to prove the performance, capacities and ability to provide the required service.

Undertake commissioning in 4 stages:

- 1 Individual component functionality testing including verification of defects free installation
- 2 Major items of plant and equipment and components.
- 3 Systems and sub-systems.
- 4 Inter-system operation and interfaces.

Commissioning to include:

- 1 Performance of individual plant items and components.
- 2 Operating sequences, interlocks and safeties.
- 3 Final controls calibration.
- 4 System operation under all operating modes and under all conditions of load.
- 5 Inter-system operation and correct interfacing connections under all operating conditions and under simulated fire conditions.
- 6 Noise and vibration tests.
- 7 Thermoscan survey of switchboards.
- 8 Rectification and correction of any defects and deficiencies.
- 9 Continue commissioning until achievement of correct performance and operation.
- 10 Electrical testing and verification to AS/NZS 3000.
- 11 Cleaning of plant room, switchboards and all equipment and devices.

6.5 FINAL CONTRACT AND HANDOVER DOCUMENTATION

6.5.1 GENERAL REQUIREMENTS

- 1 Documentation is required in electronic and hard copy format. Electronic documentation shall be supplied on the latest Microsoft and AutoCAD Software versions at the time of Practical Completion.
- 2 Scope of the documentation and format requirements comprise:

DOCUMENT	FORMAT	SETS
Operating and Maintenance Manuals, including Final commissioning data	Electronic installed on BMS File Server	1
	Electronic Storage Device – Microsoft Office Latest Version	3
	Hardcopy	3
As Built Drawings	Electronic Storage Device – AutoCAD	3
	Hardcopy – Full Size with hanging strips	1
	A3 Size	2
OH&S Safety Signage	Permanent	1

6.5.2 AS BUILT DRAWINGS

- 1 All As Built/As Installed documents submitted shall have a stamp with similar words to the following and be signed prior to submission for review.

AS INSTALLED DRAWING
We certify that this drawing is an accurate record of the work installed by our company under the terms of our contract.
Contractor's Name
Contractor's Address
.....
Signed Date
Printed

- 2 Progressively record changes to form a record of work as installed.
- 3 Prepare and submit for approval as-built drawings covering all aspects of the work.
- 4 Show the 'as-installed' locations of building elements, plant and equipment with particular emphasis on items requiring maintenance or clearing.
- 5 Provide As Built documentation as per Schedule of Required As Builts:

6.5.3 OPERATING AND MAINTENANCE MANUALS

- 1 Provide comprehensive Operating and Maintenance documentation to cover all installations in order to provide a detailed understanding of the plant and its operation, an aid for training of operators, a reference for fault diagnosis and a framework for preventative and breakdown maintenance.
- 2 Submit manuals in two stages. Stage 1 include all of the sections (generally sections 1-5) of manuals excluding final testing and commissioning records and As Built drawings. This manual shall be submitted well in advance of the Practical Completion and should be signed off by the Consultant Engineer at least 4-6 weeks before programme Practical Completion Date.
- 3 First draft manuals: Submit a first draft manual 8-10 weeks before the date for practical completion for review.
- 4 Second drafts: Following receipt of comment on the first draft and not later than 2 weeks after submitting a completed second draft for review. Incorporate comments.
- 5 Third Draft: Include all remaining sections and submit 2-3 weeks before Practical Completion. Incorporate all changes and resubmit until all comments are addressed to the satisfaction of the Consultant Engineer.
- 6 Final copies: Submit 3 sets of final volumes within 2 weeks after practical completion. Incorporate feedback from review and from training of Principal's staff, including preparation and insertion of additional data. Include a section containing commissioning test reports.
- 7 Revisions: Prior to final completion, submit 3 sections sets of loose leaf amendments for insertion in the manuals, incorporating feedback from the maintenance period.
- 8 Warning and Cautions: Include to emphasise conditions hazardous to personnel or equipment, giving instructions to avoid the hazard. Format to be:
- 9 **WARNING:** An examining or testing procedure or practice which must be observed or risk loss of life or injury to personnel.
- 10 **CAUTION:** An examining or testing procedure or practice which must be followed or risk damage to equipment.
- 11 The manual shall be neatly prepared and bound in one, or a series of, vinyl hard-back folder with lettering on the front and the spine including the following information:

OPERATION AND MAINTENANCE MANUAL

VOLUME NUMBER

CQU Rockhampton

DATE OF ISSUE

VERTICAL TRANSPORTATION SERVICES

CONTRACTOR

CONSULTANT ENGINEERS – WSP

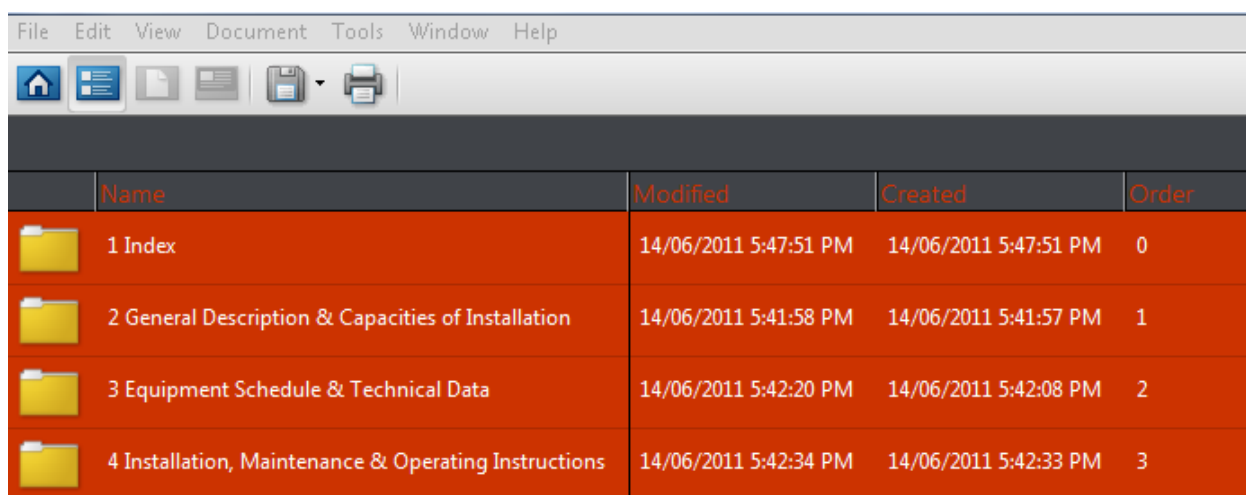
SUB CONTRACTOR

- 12 Pagination: Number pages consecutively. Double sided on A4 size (minimum 50%) recycled white (minimum CIE whiteness of 140) paper, with reinforced binder perforations.
- 13 Ring size: 50mm maximum, with compressor bars.

- 14 Dividers: Durable divider for each separate element, with typed description of system and major equipment components. Clearly print short titles under laminated plastic tabs.
- 15 Drawings: Fold illustrative drawings to A4 size and accommodate them in plastic pockets in the binder. Title block to be visible without removal of drawing
- 16 All aspects of the style and quality of the manual, including folders and contents shall be submitted for approval prior to completion.
- 17 Allow for loading the manual onto the Building BMS systems (where available) and arrange to provide the manual including commissioning results.

ELECTRONIC/DIGITAL FORMAT

- 1 Documents shall be stored in DVD Disc or Potable USB Flash Drive Storage Drive. Provide labelling to the storage media.
- 2 Electronic Storage device shall not be “copy protected”.
- 3 Manual shall be contained within **a single searchable PDF** accessible via Adobe or similar PDF reader. PDF shall be indexed, divided in to folders and titled. Each folder shall contain relevant electronic/digital format documentation in line with the contents specified below. Refer below of a screen print of Adobe Reader for a typical arrangement.



	Name	Modified	Created	Order
	1 Index	14/06/2011 5:47:51 PM	14/06/2011 5:47:51 PM	0
	2 General Description & Capacities of Installation	14/06/2011 5:41:58 PM	14/06/2011 5:41:57 PM	1
	3 Equipment Schedule & Technical Data	14/06/2011 5:42:20 PM	14/06/2011 5:42:08 PM	2
	4 Installation, Maintenance & Operating Instructions	14/06/2011 5:42:34 PM	14/06/2011 5:42:33 PM	3

“Screen Shot” for Reference

- 4 Avoid submitting manually scanned Manufacturers’ data, where possible include only searchable PDF documentation.
- 5 Include only relevant diagrams, equipment data in clear concise English. Avoid jargon and esoteric pseudonyms.
- 6 Provide each document included in the PDF with the following naming convention:
- 7 Section “X” – “Document Description”.pdf
- 8 Provide clear scanned copies of Statutory Certificates of Compliance

- a** Accessible Dwg format CAD drawings of the as-built documentation shall be contained within the PDF.

OPERATION AND MAINTENANCE MANUAL FORMAT

1 Section 1 – Index

- a** Index all sub-divisions of each section including lists of drawings, equipment, etc. for quick reference.
- b** Include index in every volume of the manuals.

2 Section 2 - General Description and Capacities of Installations:

- a** Divide into sub-sections for general items and for each individual system as appropriate and include full details
- b** Include a directory: Names, addresses and telephone, email and facsimile numbers of Principal, Consultant Engineer, Sub-Consultants, Contractor and names of responsible parties.
- c** Include a scope of works: Statement of scope of services and interfacing with other Contracts.
- d** Include an installation description: General description of installation.
- e** Systems description: Technical description of the systems installed, written to ensure that the Principal's staff fully understand the scope and facilities provided. Identify function, normal operating characteristics and limiting conditions. Include schematic diagrams.
- f** System performance: Technical description of the modes of operation of the systems installed.

3 Section 3 – Equipment:

- a** Sub-divide as for Section 2 and list all major items of equipment installed complete with manufacturer's name, agent's name, contact details, model and/or type No., serial No., size, design ratings and including all relevant data necessary for re-ordering or replacing the equipment or components of same. As far as practicable, all equipment should be broken down to individually identifiable items.
- b** Provide manufacturers' technical literature for equipment installed, assembled specifically for the project, excluding irrelevant matter. Mark each product data sheet to clearly identify specific products and component parts used in the installation and data applicable to the installation. Do not include advertising literature. Cross out any section of literature which is not applicable to the project.
- c** Include descriptions of operating systems, and major hardware such as safety gear, motor, controller, PDO, light guardian, direct drive, ropes, remote monitoring system), hydraulic equipment, etc.
- d** Include a detailed description of special operating modes as may be specified, examples being priority service, fire recall, standby power operation, Utility lift modes, and the access control interfaces complete with detailed schematics.

4 Section 4 – Installation, Maintenance and Operating Instructions:

- a** Sub-divide as for Section 2 and include manufacturer's installation, maintenance and operating instructions for each item of equipment.
- b** Emergency maintenance procedures, including telephone numbers for emergency services and after hour's contact for suppliers and Contractors and procedures for fault finding.
- c** Manufacturer's technical literature as appropriate. Register with manufacturer as necessary. Retain copies delivered with equipment.
- d** Detailed recommendations for preventative maintenance frequency and procedures which should be adopted by the Principal to ensure the most efficient operation of the systems installed. Include inspection, testing and maintenance programme in tabular form showing frequency and level of routine checks for each item.
- e** Safe trouble shooting, disassembly, repair and reassembly, cleaning, alignment and adjustment, balancing and checking procedures. Provide logical step-by-step sequence of instructions for each procedure. Include calibration and recommissioning of controls.
- f** Schedule of spares recommended to be held on site, being those items subject to wear or deterioration and which may involve the Principal in extended deliveries when replacements are required. Include complete nomenclature and model numbers, and local sources of supply. Include sectionalised diagrams of machines identifying component parts.
- g** Schedule of normal consumable items, local sources of supply and expected replacement intervals up to a running time of 40,000 hours. Include lubricant and lubrication schedules for equipment where applicable.
- h** Add to the manufacturer's technical literature, for assembly, operating, maintenance and disassembly, safety information for all plant, systems or structures installed within the project or for existing plant, systems or structures where assembly, operating, maintenance and disassembly are altered or amended by the installation of plant, systems or structures within the project.
- i** The added safety information shall be included at relevant sections of the manufacturer's literature and not as a separate part and shall be made clear and obvious to the user of the manual.

5 Section 5 – Plant Operating Instructions:

- a** Sub-divide as for Section 2 and provide a complete description and correct sequence of all actions necessary to start-up, operate and shut-down all plant including procedures for seasonal changeovers where applicable. These instructions shall include full information on such items as normal and abnormal gauge readings, instrument settings and control points, differentials, time delays, oil levels, water temperatures, and all similar relevant variable and adjustable items, to permit checking and adjustments where practicable and identification of hazardous conditions or malfunction of plant. In addition, the instructions shall include information on the immediate action to be taken in the event of hazardous conditions arising.
- b** Add to the manufacturer's technical literature, for assembly, operating, maintenance and disassembly, safety information for all plant, systems or structures installed within the project or for existing plant, systems or structures where assembly, operating, maintenance and disassembly are altered or amended by the installation of plant, systems or structures within the project.

- c** The added safety information shall be included at relevant sections of the manufacturer's literature and not as a separate part and shall be made clear and obvious to the user of the manual.
 - 6** Section 6 – Performance Test Results:
 - a** This section shall be sub-divided as for Section 2 and shall include all performance test results as outlined in previous Clauses.
 - 7** Section 7 – Certificates & Warranties
 - a** Statutory Certificates of Compliance for:
 - b** Electrical work.
 - c** Copies of manufacturers' warranties.
 - d** Certificates from authorities and utilities.
 - e** Production certification.
 - f** Copies of test certificates for the installation and equipment used in the installation.
 - g** Product type test certificates.
 - h** Electrical certificates of compliance.
 - i** Inspector's compliance certificate.
 - j** Plant design notification and registration documents as required by the Occupational Health and Safety Act and Plant Safety Regulations.
 - k** A statement of the Codes and Standards with which the installation complies.
 - l** Hazard and risk assessment report.
 - m** Acoustic and vibration report.
 - 8** Section 8 – Drawings
 - a** Drawings and technical data: As necessary for the efficient operation and maintenance of the installation.
-

6.5.4 WARRANTIES

- 1** Warranty periods are deemed to end at expiry of defects liability period unless specified otherwise.
- 2** Name the Principal as warrantee. Register with manufacturers as necessary. Retain copies delivered with components and equipment.
- 3** Commence warranty periods at practical completion or at acceptance of installation, if acceptance is not concurrent with Practical Completion.
- 4** If installation is not being manufactured and product warranty is conditional on the manufacturer's approval of the Contractor, submit the manufacturer's written approval of the installing firm.

6.5.5 REQUIRED OPERATING INSTRUCTIONS

- 1 Provide operating instructions and schematic diagrams mounted alongside equipment to meet statutory requirements, e.g. Fire Brigade. Diagrams to be colour-coded, colour-fast, laminated and mounted.

6.5.6 TRAINING OF PRINCIPAL'S NOMINATED PERSONNEL

- 1 The service of a fully knowledgeable Commissioning Engineer shall be required for a period of 2 full working hours to instruct the Owner's nominated representatives in all details of the plant operation.
- 2 Conduct training at agreed time, at system or equipment location.
- 3 Immediately after Practical Completion, explain and demonstrate to the building users or nominated representative, the purpose, function and maintenance of the installations.
- 4 Use qualified personnel who are knowledgeable about the installations.
- 5 Prior to Practical Completion, explain and demonstrate to the Principal's nominated personnel the purpose, function, operating and maintenance of the installations.
- 6 Use items and procedures listed in the Operation and Maintenance manuals as the basis for detailed instruction of Principal's nominated personnel.
- 7 Conduct training at agreed time, at each system of equipment location.
- 8 Upon completion, submit certificates of training, signed by trainer and trainees.
- 9 Provide qualified manufacturer's training representatives who are knowledgeable about the installations
- 10 The services of the Contractor's Commissioning Engineer and BMS controls commissioning engineer shall be required for a period of 8 weeks after Practical Completion to assist the operator in the operation of the Facility.
- 11 During the warranty period, provide technical assistance and advice to the Principal's nominated personnel regarding the operation and maintenance of the plant.
- 12 For equipment requiring seasonal operation, demonstrate during the appropriate season and within 6 months of Practical Completion.
- 13 Include copy of training material within the O&M Manual

6.5.7 SPARE PARTS SCHEDULE

At least 8 weeks before the date for Practical Completion, submit a schedule of spare parts necessary for maintenance of the installation. State against each item the recommended quantity, and the manufacturer's current price, including for

- 1 Packaging and delivery to site;
- 2 Checking receipt, marking and numbering in accordance with the spare parts schedule;
- 3 Referencing equipment schedules in the operation and maintenance manual; and

6.6 POST PRACTICAL COMPLETION AND DEFECTS LIABILITY PERIOD

6.6.1 WARRANTY & DEFECTS LIABILITY

Warrant the whole of the Contract Works, for the Defects Liability Period (DLP), against defective workmanship and materials and against non-compliance of equipment and/or complete system with specified performance and operation.

The Defects Liability Period shall continue for a period of 12 months after the date of issue of Practical Completion and during this period, the Contractor shall be responsible for making good on a timely manner, defects arising from the defective design materials, premature component failures, or workmanship or from any act of the Contractor that may develop in the work.

Perform maintenance during the warranty period to maintain warranty conditions. Maintenance shall comply with the current regulatory requirements.

In the event of inclusion of equipment normally covered by a lesser time warranty, or whose warranty would otherwise normally expire during the warranty period, allow for and include the costs of extending such warranty to that covering up to the end of the Defect Liability Period.

Where warranted work or equipment has failed, the warranty shall recommence on the date of acceptance of the rectified work for an additional period of time, extended by the period during which the Principal was without the use of the work or equipment. For rectified work involving new replaced equipment, the warranty for the new component shall be from date of replacement for the 12 months or longer as required by this specification

6.6.2 GENERAL DLP MAINTENANCE SCOPE

Maintenance services shall include:

- 1 Preparation of a maintenance management plan which includes a detailed maintenance programme, equipment schedules, statutory and preventative maintenance schedules, OH&S risk assessment, inspection and testing schedules, breakdown rectification management and reporting systems. Submit maintenance plan and schedules prior to Practical Completion
- 2 Preventative and corrective maintenance and testing services to assure specified system reliability and availability.
- 3 Statutory maintenance and testing required by the Building Regulations and other authorities.
- 4 Breakdown and system fault responses. Attend site, ascertain cause of alarm and rectify where possible. Make system alterations at own cost to prevent future system faults. Pay any Authority charges resulting from callout on system faults.
- 5 Statutory regular system fire and life safety system control and performance verification test to be carried out after hours.

- 6 Comply with Building and Occupational Health and Safety Regulations Comply with manufacturer's recommended maintenance programme and schedules.
 - 7 Monthly maintenance and servicing in accordance with maintenance schedules recommended by equipment manufacturers.
 - 8 Rectification of defects at Contractors cost. Replacement of consumables at Contractors cost during the maintenance period.
 - a Submission of monthly reports to Principal and Consultant Engineer on maintenance activities performed.
 - b Provision and filling out log book which must be kept on site, detailing all visits.
 - c Service personnel to be available for call out on a 24 hour per day, 7 days per week basis. Response time for emergency and corrective maintenance must not exceed 2 hours.
 - d Operation and verification in conjunction with other services trades, of all fire and life safety systems.
-

6.6.3 FINAL COMPLETION

Final Completion will not be granted until maintenance records are complete and signed off and until all defects have been rectified and inspected.

The Maintenance and Defects Liability Period may be extended unless maintenance and servicing visits have been made and reported as specified.

6.6.4 SUBMISSIONS

- 1 Provide;
- 2 Maintenance management plan.
- 3 Monthly Maintenance reports.
- 4 Completed maintenance records

Mandatory maintenance records to include monthly maintenance and service reports, reports on all defects and their rectification, certification of the correct operation of essential services, a maintenance log book and any other records which must be kept in order to comply with legislative or OH&S requirements.

Monthly Maintenance and Service Reports to include:

- 1 Check list of all items serviced and inspected.
- 2 Description of maintenance performed.
- 3 Notes on supplementary maintenance required.

Maintenance Log Book to include:

- 1 Schedule of all equipment.
- 2 Register of maintenance visits and cross reference to service reports.

- 3 Register of work performed under headings of preventive maintenance, corrective and emergency maintenance and defects rectification.

The Maintenance Log Book is to be kept in a locked cabinet provided for the purpose in an agreed location on site.

At the conclusion of each maintenance visit the log book is to be duly completed and signed off by an authorised representative of the Principal /Building Owner and service records are to be forwarded to the Principal /Building Owner or his representative within 3 days of each visit.

6.6.5 DLP PREVENTATIVE MAINTENANCE AND TESTING

GENERAL

As a minimum, carry out all maintenance and servicing recommended by the manufacturer of each piece of equipment at the recommended intervals.

SCOPE

Defects Liability Maintenance is included in the Works.

GENERAL REQUIREMENT

Carry out Defects Liability Maintenance to the end of the Defects Liability Period using adequate number of competent maintenance personnel, providing sufficient spare parts as required to maintain the Works with minimum down time servicing and maintaining the Works.

PARTICULAR REQUIREMENTS

Carrying out comprehensive maintenance adequately and periodically in accordance with following requirements.

- 1 Maintenance of the Works in condition which is safe, complying, reliable and in good state of cleanliness and repair;
- 2 Regularly servicing the Equipment with visits at least twice monthly;
- 3 Provision of a 24-hour call-out service with call-outs attended within guaranteed maximum response time, and without charge;
- 4 Restricting the average number of call-outs for each Unit, excluding nuisance calls, to nine per year;
- 5 Maintenance of performance requirements established for the Works;
- 6 Completion within one month of date of Practical Completion of defects and work outstanding at date of Practical Completion;
- 7 Conducting tests of the installation in accordance with the requirements of the manufacturer's requirements; the relevant design standard, and include the following:
 - a Annual governor, safety gear and buffer tests;
 - b Monthly phone system and alarm device test;

- c** Quarterly emergency light test;
 - d** Annual ride quality and lift tune measurement, and report to the building owner.
- 8** Removal of minor rubbish from the pit during regular service visits;
- 9** Replacement of lamps and tubes installed in the Equipment during regular service visits;
- 10** Keeping records relating to service visits, call outs, faults rectified and repairs carried out;
- 11** Provision of a statement detailing such information when instructed;
- 12** Completion of an on-site log book recording attendances and nature of the work carried out;
- 13** Maintenance and update of essential data Including wiring diagrams, drawings and software listings necessary to maintain the Works;
- 14** Immediate replacement of Equipment which fails as result of incorrect servicing, faulty workmanship, materials or design;
- 15** Replacement of failed Equipment manufactured overseas using air-freight if not in stock;
- 16** Replacement of faulty Equipment outside normal working hours of premises if so instructed.

7 TECHNICAL REQUIREMENTS – LIFTS

7.1 CONTROL SYSTEM

7.1.1 GENERAL CONTROL SYSTEMS FEATURES

17 As a minimum, all lifts shall have at least the following features:

- a** Exclusive service.
 - b** Fireman's service.
 - c** Load-weighing control.
 - d** Door nudging system.
 - e** Anti-nuisance feature.
 - f** Emergency Power operation.
 - g** Automated switching of light fittings in lift cars.
-

7.1.2 GENERAL/CONVENTIONAL CONTROL SYSTEMS

- 1 The lift control systems shall be high speed micro-processor and software based and incorporate the latest proven demand based traffic management algorithms to optimise system response times and operating/energy efficiencies.
- 2 Provide high speed microprocessor based control systems compatible with the types of equipment specified and commensurate with the efficient operation of the building.
- 3 The control system shall be able to utilise neural network topology to dynamically adapt to passenger traffic demand.
- 4 The control system shall incorporate facilities to manage long wait calls, respond to dynamically changing car loading, and respond to learned traffic patterns.
- 5 The power control system shall be the solid state type that controls and varies the voltage supply to the lift motor to provide stepless acceleration, smooth deceleration and a soft final stop.
- 6 The system shall control acceleration, deceleration and full speed running to preset patterns. The lifts shall restart automatically on resumption of power supply after a power failure. A master controller shall start the lifts in a sequential order.

7.1.3 SPEED CONTROL SYSTEM

- 1 Closed loop control equipment operating on distance measuring principle.
- 2 Smooth step-less acceleration and deceleration, compensated to minimise fluctuations in acceleration and deceleration and full speed rate which may be influenced by variations such as load, temperature and humidity.
- 3 Final stop by electrical control of motor.
- 4 Brake application after lift stopped and at floor level.
- 5 Final stop with levelling accuracy of ± 10 mm under all load conditions.
- 6 The speed control system and motor drive shall incorporate a 'full scale' lift position detection or measurement system that enables the lift to be consistently stopped at floor level within the accuracy nominated using closed loop control principals. The system shall re-level to floor level to compensate for rope stretch as the load changes when passengers/load are added or removed from the lift at floor level. Movement of the lift at floor level with the doors open shall be provided and shall be carried out in a smooth stepless manner that is virtually imperceptible to lift users.

7.1.4 DRIVE SYSTEM EFFICIENCY (ESD)

- 1 The lift drives shall be based on variable frequency AC permanent magnet motors.
- 2 Drives shall have a regenerative capability that recovers excess energy in an overhauling condition to be returned to the electrical mains rather than dumped as heat.

7.1.5 AUTOMATIC RESCUE DEVICE

- 1 The automatic rescue device (ARD) shall be provided on all lift cars which are not supplied with emergency backup power.
- 2 The operation of the ARD shall be as follows:
 - a The ARD shall activate in the circumstance that the lift car is travelling between floors and there is loss of mains power supply.
 - b Upon activation, the lift car shall immediately return to the closest floor at low speed.
 - c When the lift car arrives at the closest floor, it shall park and hold the doors open for a period of 30 seconds to ensure all passenger can disembark.
- 3 The lifts shall return to normal service automatically following the re-instatement of mains power supply.

7.1.6 ACCESS CONTROL

- 1 All the control systems shall interface with the access control system as may be specified elsewhere.

- 2 The lift control system shall interface with a security management system. When a signal is received from the system that the security is on, car calls to all or any floors selected by the security system may not be registered.
 - 3 Car calls to the secured floors may be registered by insertion/wiping of a coded card key which will permit certain call/s to be registered.
 - 4 Interfacing shall be provided between the lift control system and the security control system.
 - 5 The interface level shall be as per that specified elsewhere.
 - 6 Supply feedback signals to the security controller indicating which floor button has been selected by the card user.
 - 7 When in the secured mode, the group control system shall not permit dispatching of lifts to a parking floor if that floor is on security.
 - 8 Allow for all wiring between the interface panel connected to by the Security Services Contractor and lift controllers and from lift controllers to each card reader
 - 9 Allow space in all car operating panels for installation of card readers. Card readers shall be housed behind a Perspex window. Install card readers supplied by Security Subcontractor.
-

7.1.7 FLOOR LEVELLING

- 1 Floor levelling shall be obtained in a stepless manner direct to the floor to within +/- 10mm under all load conditions.
 - 2 The levelling equipment shall be silent in operation and of a type which does not use mechanically operated switches in the well.
 - 3 A low speed re-levelling feature shall be provided to maintain correct level if for any reason level within guaranteed tolerances is not achieved in the initial approach to floor or while loading or unloading at floor level.
-

7.1.8 EXCLUSIVE SERVICE

- 1 Operation of the exclusive service switch will cause the lift to be removed from normal service and respond to car calls only. If it is a single lift then all landing call buttons will become inoperative and any registered hall calls will be cancelled. Hall lanterns shall not operate while the lift is on independent service.
- 2 The use of the exclusive service switch provided in one car operating panel shall:
 - a remove the lift from automatic control whereby it does not respond to or cancel registered landing calls
 - b park the lift with doors open
 - c open panel to show hidden car call buttons
 - d allow car button registration to be made in the normal manner and with respect to further travel of the lift, shall set the direction of travel

- e constant pressure on the door close button shall cause the doors to close and the lift to respond to a registered car call, all car calls shall be cancelled upon opening or reopening of the doors
 - f cause the door passenger protection device to become inoperative
 - g not cancel car button security where provided.
-

7.1.9 FIRE SERVICE

- 1 Fireman's Service shall be in accordance with BCA requirements
-

7.1.10 LOAD WEIGHING

- 1 A load weighing feature of the strain gauge, load cell or other appropriate type, shall be provided to achieve the following:
 - a When loaded in excess of 110% of rated capacity the lift shall be prevented from leaving that floor. The doors shall remain open, a warning buzzer shall sound and the lift overloaded indicator shall illuminate until the load is reduced to below this capacity.
 - b When loaded in excess of 80% of rated capacity the lift shall bypass landing calls. The bypass setting shall be readily adjustable between 50% and 80%.
 - c When no load is detected in the lift car, all car calls previously registered shall be cancelled.
-

7.1.11 DOOR NUDGING

- 1 The passenger protection device shall incorporate a nudging feature, which, after an adjustable pre-set time, shall cause the doors to close at a reduced speed and sound an audible buzzer. Such nudging feature to be initiated under any of the following conditions:
 - a Obstruction of the passenger protection device for an extended period.
 - b The passenger protection device becoming inoperative for any reason.
 - c Stuck landing button at a floor where the lift is parked.
 - d The door open button shall remain operative at all times.
-

7.1.12 ANTI-NUISANCE

- 1 Anti-nuisance feature, designed to cancel registered car calls in circumstances where the number of car calls exceeds the probable number of people in the lift car shall be provided.
-

7.1.13 ENERGY SAVING FEATURES

- 1 Automated switching of light fittings and display screens in lift cars shall be provided, to reduce the electrical load to the minimum allowable when lifts are idle. Lights shall automatically be switched off whenever the lift has been idle for 2 minutes, except when the lift is in a special operation mode (exclusive, fire, etc.), or if the lift is in a failed start or fault condition.

7.1.14 INSPECTION CONTROL

- 1 Provide top of car operating devices on the roof of each lift car featuring the following controls:
 - a Constant pressure up/down and common buttons.
 - b Door open and door close switches.
 - c Emergency stop switch.
 - d Switch removing car from normal operation ready for inspection operation.

7.2 SPECIAL CONTROL SYSTEMS – NTB LIFT ONLY

7.2.1 HAZARDOUS GOODS TRANSPORT MODE SERVICE

- 1 All landing panels (LOP) for goods lifts will be provided with a three position key operated switch labelled “HAZARDOUS GOODS OPERATION” with the positions labelled as follows:
 - a Close doors
 - b Off
 - c Call lift
- 2 The lock will be spring return to the “OFF” position from both other positions.
- 3 In addition to normal switches, there will be a two-position switch in the car operating panel (COP) labelled “HAZARDOUS GOODS OPERATION”. The two positions will be labelled:
 - a On
 - b Offand the key can be withdrawn in either position.
- 4 The key switches in both the COP and the LOP will be of the Bi-Lock type.
- 5 When the hazardous goods service (HGS) key switch is in the “OFF” position the designated lift will operate normally
- 6 The attendant turns the key switch in the landing operating panel (LOP) clockwise from the “OFF” to the “CALL LIFT” position.
- 7 An in-car announcement is made: “Please exit at your destination stop, this lift is required for a special service and will not take new calls”. Note, this audio announcement will repeat approximately every 10 seconds
- 8 An illuminated flashing sign in the lift COP will light “Special service operation”
- 9 Landing call response is inhibited

- 10 The lift will complete all pre-registered car calls, all new calls will not be accepted. After all passengers leave the lift car, the doors will close and the lift travel directly to answer the HSG key switch. If the lift is idle it will immediately travel directly in answer to the HSG key switch.
- 11 The lift will travel (non-Stop) to the “calling” floor (at which the HGS switch is selected.)
- 12 Open its doors.
- 13 The lift will remain at that floor with the doors open.
- 14 The attendant will remove the key switch from the landing fixture in the “OFF” position.
- 15 The lift will remain “captive” in the HGS mode of operation for 60 seconds (adjustable). If the process does not proceed to the next stage, the lift will return to normal service.
- 16 The HGS car operating panel (COP) key switch is turned to the “ON” position.
- 17 The key is removed in the “ON” position.
- 18 The attendant will ensure that no one remains inside the lift
- 19 The goods are loaded.
- 20 The key is inserted into the hall switch and turned counter clockwise to the “CLOSE DOORS” position. The doors close and the key returns to the central “OFF” position and withdrawn.
- 21 The attendant travels via other lift or stairs, to the “destination” floor.
- 22 The attendant then turns the HGS key switch in the LOP to the “CALL LIFT” position at the “destination” floor.
- 23 The lift travels to the “destination” floor.
- 24 The doors open.
- 25 The goods are removed.
- 26 The key is removed from the “destination” landing HGS key switch.
- 27 The COP HGS key switch is returned to the “OFF” position.
- 28 The key is removed.
- 29 The lift returns to normal service.
- 30 The HGS mode of operation will not initiate if:
 - a The Fire Service is operated.
 - b The lift is in Inspection mode.
 - c The lift is on Exclusive Service.
- 31 Selection of the Fire Service mode while the lift is on HGS will return the lift to Ground Floor and keep doors closed until the attendant turns the HGS key switch in the LOP to the “CALL LIFT” position and open the door for unloading
- 32 Activation of Gas Detection:

- a** Activation of Gas Detection 1- Activation of Gas Detection when lift is moving:
 - i** Lift shall stop and return to Ground Floor and keep doors closed
- b** Activation of Gas Detection 2 - Activation of Gas Detection when lift is at the destination floor:
 - i** No action, lift remain idle with doors open.
 - ii** Operator can switch the key to the “CLOSE DOORS” position from the landing
 - iii** Once the doors are closed, the lift shall return to Ground Floor and keep doors closed
 - iv** The attendant travels via other lift or stairs to Ground Floor.
 - v** The attendant turns the HGS key switch in the LOP to the “CALL LIFT” position at Ground Floor to open the doors. Interrupt Feature

- 1 The Equipment shall not contain any Interrupt Feature.
- 2 “Interrupt Feature” (definition) – an internal feature or device within the Equipment or an external input to the Equipment that has the result of interrupting the complete and normal automatic operation of the Equipment and its maintenance and adjustment, either based upon the passage of time or number of operational sequences as for example lift starts or any other predefined or random number or sequences of events, where the restoration of the complete and normal automatic operation of the Equipment and its maintenance and adjustment requires the application of some device or input or procedure.
- 3 Excluded from this definition are only those necessary Lift Code related fail-safe devices that temporarily stop the Equipment from operating as the result of foreseeable Equipment failure or malfunction, where the restoration of normal operation is effected by correcting the malfunction.

7.3 KEY SWITCHES AND KEYS

- 1 Keys switches shall be of a highly restricted type for: Master keys, Car light key switches, Fire Service, and Exclusive Service key switches, and other such keys as may be required.
- 2 Supply three sets of standard regulatory keys for recall fire control and exclusive service.
- 3 Supply one set of standard liftwell unlocking devices and keys as required in each controller cabinet in the case of MRL lifts.
- 4 Supply two sets of keys for key boxes as may be fitted.
- 5 Supply the above specified number of sets of suitably tagged and labelled keys at Practical Completion.
- 6 Note special Regulatory key requirements for recall fire service control switch
- 7 Fire service key shall be kept at the FIP (Fire Indicator Panel)

7.4 APPOINTMENTS

7.4.1 ENGRAVING

- 1 Engraving in stainless steel faceplates shall be of adequate depth for adhesion and filled with epoxy resin, as appropriate, that is rendered flush during the faceplate finish, or an alternative fill treatment to approval that provides an equivalent permanency and durability. Paint is unacceptable.
- 2 Colours of infill: generally black or white; red where Code requires same; white for alarm/emergency telephone.
- 3 BCA signage regarding use of lifts in fire shall be engraved on the faceplate. The use of additional signage which is adhered to the lobby wall is not considered acceptable.

7.4.2 FIXINGS

- 1 This section is applicable to car operating panels, hall lanterns and hall button stations:
 - a Fixings shall permit alignment of faceplate and removal without damaging adjacent wall finish.
 - b Minimum of two fixings per faceplate.
 - c Adequate fixings to hold faceplate flat to adjacent wall finish.

7.4.3 LANDING CALL BUTTON STATIONS

- 1 Servicing and part replacement able to be readily done from lobby.
- 2 Removal of faceplate readily achievable without disturbing or damaging adjacent wall finish.
- 3 Alignment shall be parallel with landing frame and adjustable. Height above floor to suit disabled requirements.
- 4 Faceplate shall be engraved regarding use of lift in fire to Code requirements.
- 5 Incorporate access control card reader supplied where appropriate.
- 6 Incorporate Fire Service switch where required as required by regulation
- 7 Securely fixed as specified elsewhere.
- 8 Mandatory fire signage "DO NOT USE LIFTS IF THERE IS A FIRE" needs to be included at the Landing Call Button Stations

7.4.4 LANDING INDICATORS

- 1 One per lift entrance at location to be advised either beside the architrave or over it.
- 2 Servicing and part replacement able to be readily done from lobby.
- 3 Removal of lens and faceplate readily achievable without disturbing or damaging adjacent wall finish.

- 4 Alignment shall be parallel with landing frame and adjustable.
 - 5 It shall indicate direction of travel of arriving lift by visible and audible means.
 - 6 Terminal floors shall only have a single arrows indicating in the only possible direction of travel.
 - 7 It shall incorporate a device comprising electronic gong or chime located within lantern box to produce sound signal.
 - 8 Two sound signals for downward travel of arriving lift and one sound signal for upward travel.
 - 9 Sound signal devices shall be individually adjustable in sound output level.
 - 10 Gong or chime shall be capable of being switched off by the activation of a signal from a time clock or a signal from the access control system.
 - 11 Lens illumination shall be adequate and white and evenly distributed without bright spots.
 - 12 Light leakage shall be eliminated.
 - 13 Securely fixed as specified elsewhere.
-

7.4.5 CAR OPERATING PANELS

- 1 All lifts shall be provided with car operating panel(s) incorporating the following:

MAIN CAR OPERATING PANEL

- 1 The Main Car Operating panel shall generally contain:
 - a Surface engraving describing functions. All functions shall be described in words as well as mandatory symbols.
 - b Floor call buttons.
 - c Door open and close buttons labelled “Door open” and “Door close” respectively.
 - d Access control card reader if applicable.
 - e Alarm and emergency communication button.
 - f Communication speaker hole approximately at 1,500 above floor level.
 - g Communication loudspeaker behind engraved holes and above loudspeaker level.
 - h Key switches.
 - i Surface engraved regulation load notice.
 - j Signage including lift reference number, year of manufacture and other code related requirements
 - k Surface engraved notice giving lift and building location.
 - l Car information display if not located centrally over entrance.
 - m Door hold button as may be specified elsewhere.

AUXILIARY CAR OPERATING PANEL

- 1 The Auxiliary Car Operating Panel shall generally contain:
- 2 All items as noted for the Main Car Operating Panel with the exception of the key switches.

LAYOUT PRINCIPALS

- 1 The layout of the car operating panel(s) shall generally be as follows:
 - a Full height (2,100h min) car operating panels to be provided.
 - b As simple and uncluttered appearance as practicable.
 - c Controls grouped to facilitate easy and ready recognition of functions.
 - d Related functions grouped together, and other functions located apart.
 - e Main functions of floor and door buttons grouped together.
 - f Floor buttons arranged in a single row unless prevented by requirements for access by the disabled.
 - g Necessary multiple rows of floor buttons arranged in “zig-zag” sequence rather than sequentially up each column then starting up the adjacent column.
 - h The car operating panel shall be sufficiently wide to allow all call buttons to be located within the height zone required by AS1735.12 for access by people with disabilities.
 - i Door open button closer to entrance side; door close button on other side.
 - j Alarm/emergency communication button the only control at that height if layout permits, located in bottom right hand corner as per AS1735.12:2020.
 - k Access control card reader below floor buttons.
 - l Key switches shall be located well above or below general use functions.

CAR OPERATING PANEL CONSTRUCTION

- 1 Car operating panel faceplates shall be a minimum of 2.0 mm thick and shall be of stable construction and visually flat without perceptible deflection. Car operating panels shall sit flush against the car wall and not project past the lift wall finishes and shall be of the hinged type.

7.4.6 CAR INFORMATION DISPLAYS

- 1 A car information display shall be provided in each car operating panel and shall be mounted at a height of not less than 1,600mm above the car floor. The display shall be located between 1,600 and 1,800 above the car floor.
- 2 Car information displays shall consist of the following as a minimum:

STANDARD DISPLAYS

- 1 The display shall employ LCD digital position and direction of travel indication to approval with a minimum character height of 50mm.

- 2 The display shall display messages dependant on the lift status Including: name of floor at which lift is arriving, “Exclusive Service”, “Fire Service”, “Out of Service”, other fault/status messages as appropriate.

7.5 LIFT CAR

7.5.1 CAR PLATFORM

- 1 The car platform shall be designed and robustly constructed to cope adequately with all possible conditions of use to which a lift of its type could reasonably be subjected to.
- 2 The platform surface shall be smooth, free from flexing or drumming and isolated from the main car frame structure by purpose designed sound and vibration isolation mountings.
- 3 The front edge of the platform shall be fitted with extruded hard aluminium alloy sill together with a metal toe guard. Stainless steel may be used as an alternative.
- 4 The underside of platforms shall comply with fire proofing requirements.

7.5.2 CAR FRAME CONSTRUCTION

- 1 Car frames shall be robustly constructed of structural steel members to suit the specified duty and to form a rigid structure.
- 2 The spacing between top and bottom cross members/channels shall be the maximum permissible within the specified overrun dimensions to ensure maximum vertical spacing of the car roller guides. Care should be taken to ensure roller guide centres do not coincide with rail fixing centres or rail joint centres.
- 3 Where 2:1 roping is used, car sheaves shall be enclosed, acoustically treated and effectively isolated from the car frame.
- 4 All car hoist and/or compensation rope/chain hitches shall be damped and suitable isolating pads fitted to prevent transmission of noise/vibration to car frame.
- 5 Compensation ropes/whisperflex where used, shall be effectively isolated to minimise transmission of sound and vibration and terminated at the centre of gravity on the lift car and near the top of counterweights.
- 6 Lift cars shall be statically balanced with particular attention to location of car pick up, arrangement and location of trailing flexes and roof mounted equipment.
- 7 Counterweights shall also be fully balanced after correct final weights are established.

7.5.3 CAR GUIDE SHOES

- 1 The car frame shall be fitted with guide shoes, spring loaded and adjustable to ensure sufficient contact with the guide rails.

- 2 Guide shoes are to be provided for all cars and counterweights. They shall be rubber or polyurethane tyred and shall be smooth and silent in operation.
 - 3 Guide shoes shall be located at maximum vertical spacing permissible within the specified overrun dimensions. Care should be taken to ensure roller guide centres do not coincide with rail fixing centres or rail joint centres.
 - 4 For speeds up to 1.6m/s sliding shoes are acceptable, for speeds above 1.6m/s and less than 3.0m/s roller guides shall be provided.
 - 5 All roller guides shall be fitted with high quality bearings.
 - 6 Guide assemblies shall be effectively vibration isolated at point of fixing to car frames.
 - 7 Guide assemblies shall be installed strictly in accordance with manufacturer's specifications with close attention to clearances, setting of spring ratings and pressures, and effective lubrication.
 - 8 Springs shall be properly selected in relation to car mass. A small spring constant shall be used to ensure rigidity of mounting and to achieve early damping.
 - 9 Guide shoe assemblies shall be adjusted to ensure optimal ride, once fitted.
-

7.5.4 CAR SUPERSTRUCTURE

- 1 The car superstructure shall be fabricated from mild steel, suitably braced and stiffened to eliminate flexing or drumming and designed to accept removable wall panels (panels removable from within lift car).
 - 2 The car superstructure shall be effectively isolated from the car frame and shall be free of any racking, rattles or squeaks under all loading, unloading and operating modes.
 - 3 Where car superstructures comprise two shells, the outer shell shall be effectively isolated from the car frame, and the inner shell supported on anti-vibration mountings. In spaces between the inner and outer shells fit approved acoustic material to absorb noise and vibration.
 - 4 Exterior of lift cars shall feature flush surfaces to reduce turbulence and all joints shall be designed to dampen vibration and minimise noise.
 - 5 Interior finishes for lift cars shall be of the highest standards of materials and finish.
 - 6 Superstructures and finish materials shall be fully protected from when they leave the factory to when the lifts are accepted at practical completion.
 - 7 Handrails shall be provided on the top of car to meet code requirements. Handrails shall be painted yellow in colour.
-

7.5.5 CAR VENTILATION

- 1 Openings for natural ventilation shall be provided at the bottom and top of the side and rear wall panels and shall be located such that they are not directly visible from within the car.
- 2 Ventilation openings shall be fitted with baffles to minimise transmission of air borne noise.

- 3 Supply and install silent running, single phase, concealed, exhaust fans in each lift car capable of changing the air within the lift car at a minimum of 25 times per hour and having a sound level output of 45 dBA maximum measured at 1.5m above the lift car floor, inside car, doors closed, and car stationary at any floor.
- 4 Exhaust fans shall be isolated from the car structure with anti-vibration mountings.

CAR ACOUSTIC TREATMENT

- 1 Provide acoustic treatment as may be necessary to achieve the Specified acoustic performance:
- 2 Acoustic treatment shall consist of complying sound absorbing material to attenuate intrusion of mechanical and air movement noises in lift applied to outside surface of lift.
- 3 Sound absorbing material shall be selected and applied in a manner that prevents it leaving lift car surface.
- 4 It shall consist of sound absorbing material that does not allow build-up of dust on its surface and is impervious to oils and solvents.

CAR MECHANICAL VENTILATION

- 1 Car ventilation system shall consist of an extractor fan on car roof complete with sound isolated supports.
- 2 Fans shall be capable of achieving acoustic requirements.
- 3 It shall be connected to a nearby isolating switch plug.
- 4 It shall be sound isolated from lift car ceiling.
- 5 It shall extract air from lift interior without drawing air directly from adjacent natural ventilation slots at top of car walls, achieved by necessary baffling or plenum installation.

7.5.6 CAR LIGHTING AND POWER

- 1 All lighting and power shall be in accordance with AS 3000.
- 2 Where required, a single phase socket outlet with a metal fascia shall be mounted on the COP.

7.5.7 EMERGENCY LIGHTING

- 1 Each car shall be provided with an emergency lighting unit to provide internal car illumination in the event of power failure. Illumination where provided shall be included in addition to general car illumination. Preference will be given to inverter type units which illuminate a minimum of 50% of the normal car lights. The unit shall provide continuous illumination for a minimum period of two (2) hours and be completely sealed and maintenance free.
- 2 The charger shall be solid state, completely automatic, providing fast and float charge modes of operation.

7.5.8 EMERGENCY ALARM

- 1 There shall be an audible alarm on the car.
- 2 The emergency alarm shall be powered by a standby battery system.
- 3 While the alarm button is pressed, the button shall illuminate and an audible alarm shall indicate its functionality.
- 4 Ancillary alarms shall be installed within the liftwell at the main entrance level. All volumes shall be adjustable.
- 5 The alarm system shall be capable of being connected to a building management system. This connection shall be via voltage free contacts or a more advanced method (providing this method interfaces with the building management system).

7.5.9 CAR DOORS

- 1 The car doors shall be power operated, type fitted with replaceable guides designed to prevent the doors from rattling while the lift is running and to run silently in the groove of the car sill.
- 2 Car and landing doors for passenger and fire lifts are to be of the manufacturers own design but must have a minimum of two bottom shoes engaged in the track at all times. All car and landing doors shall be fitted with retaining devices such that the door cannot come out of the bottom track if all of the shoes become worn or dislodged.
- 3 Landing doors are to be fitted with adhesive floor designations numbers on the shaft side.
- 4 Car doors shall be of the horizontal sliding type designed for use with power door operator.
- 5 Doors shall open to the specified clear opening width when open.
- 6 Doors shall be aligned with adjacent car and landing entrance jambs when open.
- 7 Doors shall consist of furniture steel underlying construction suitably reinforced to achieve rigid assembly.
- 8 Doors shall incorporate removable guides on bottom edge preventing lateral oscillation and noise.
- 9 Movement and discernible noise shall be prevented while car is travelling.

7.5.10 POWER DOOR OPERATION

- 1 Door operators shall be high performance VVVF closed loop type.
- 2 Both the landing and car doors shall operate automatically and simultaneously. Pre-opening of doors may be provided.
- 3 Door opening and closing speeds and door open dwell times shall be fully adjustable and set up to meet guaranteed performance without adversely affecting passenger comfort.
- 4 Operation of the passenger protection device or the door close button shall reduce all pre-set door open dwell times to less than one second.

- 5 The whole door operation shall be accomplished smoothly and quickly without undue noise, vibration or shock. Door movements shall be cushioned or checked at both limits of travel.
 - 6 Door motor fuse and overtime protection shall be provided.
 - 7 Rated open and close cycles to accommodate specified lift run tests.
 - 8 Operate doors on car and landings quietly and smoothly.
 - 9 Allow doors to reverse or to close at reduced speed or with reduced torque at normal speed without discernible shock or movement.
 - 10 Opening, closing and dwell times shall be adjustable.
 - 11 Commence doors opening during final levelling operation of car.
 - 12 Power door operators shall incorporate closed loop feedback control allowing door torque to be sustained over the full closing cycle of the doors to avoid the impact of shaft windage and internal/external pressure differences.
 - 13 Doors shall be driven by the door operator to the fully closed position and not rely on the landing door closer to close and lock.
-

7.5.11 PASSENGER PROTECTION DEVICE

- 1 The passenger protection device shall include an illuminated indicator to visually warn passengers of doors opening and closing and include infra-red beams continually monitoring clear opening from 50mm above floor level to at least 1550 above floor level. Unaffected by dust, moisture, vibration and ambient light.
 - 2 System shall incorporate coloured indicators that operate on door movement. The detectors shall illuminate green in the doors are opening, flash red as they start to close, and stay red as the doors move together.
 - 3 The device shall function such that obstruction of the entrance shall hold doors open if in the open position, or reverse and fully reopen doors if closing.
 - 4 The device shall be positioned to avoid the risk of damage and shall not intrude on the clear door opening.
 - 5 The passenger protection device shall incorporate a nudging feature as specified elsewhere.
 - 6 Interruption of the passenger protection device shall reduce the door open dwell time to less than one second once the entrance becomes clear.
-

7.5.12 LIFT CAR PROTECTIVE QUILTING

- 1 Supply quilted canvas protective quilts of fire retardant construction complete with fixings. The fixings shall consist of plastic coated hooks securely fastened to the screen and long enough to be supported from the top of the side and rear wall panels of the respective car superstructure. Where glass panels (including mirrors) are fitted inside the lift cars, appropriate warning notices shall be displayed on the protective quilting.

- 2 Each set of blankets shall be labelled as to lift and wall of lift.
-

7.5.13 DIGITAL COMMUNICATION SYSTEM

- 1 A self-dialling hands free telephone shall be mounted in the car operating panel. The telephone shall be activated by means of pressing the phone button (to illuminate on pressing) on the car operating panel for 3 seconds, and shall automatically dial a permanently attended location. The phone system shall also be capable of receiving calls, and automatically deactivating upon time-out (adjustable), busy tone, etc.
 - 2 The system shall not require button to continue to be pressed once connection has been made.
 - 3 The system shall automatically identify to the service centre the building and lift number from which the call was made.
 - 4 A 4G dual SIM GSM digital communications platform with CANbus connection, Pixel, Avire or equivalent, capable of supporting AS1735.1, AS1735.12, AS1735.19 standards compliance, through;
 - a Monitoring, testing and reporting of emergency telephone system function, including battery backup status.
 - b Monitoring, testing and reporting of lift car light curtain function, including diode status.
 - c Monitoring the overall lift status.
 - 5 The digital communications platform shall include battery backup, self-monitoring options and compatible with all digital network phone service providers. The system shall also be capable of utilising a multi-carrier SIM.
 - 6 Should the network signal be weak, an extended antenna shall be provided to ensure a reliable network signal be provided.
 - 7 Compliance with AS1735.12 is required.
-

7.5.14 VOICE ANNUNCIATION

- 1 Voice annunciation shall be provided in each lift car to the requirements of AS1735.12:2020.
 - 2 The volume of the voice annunciation shall with adjustable and the voice projection shall be clear and free from accent. The voice characteristics shall be to approval.
-

7.5.15 PUBLIC ADDRESS AND HEARING AUGMENTATION

- 1 Separate 100V public address (PA) speaker, lift ceiling hearing augmentation loop and Ampetronic model ILD15BB (or equivalent) loop driver shall be provided on the lift car.
- 2 The hearing augmentation loop driver shall repeat all PA announcements made to its respective station platform from the operating centre and all lift car level annunciations heard on the lift car level annunciator speaker.
- 3 The recommended Ampetronic ILD15BB loop driver includes a high impedance 100 V line audio input, a low impedance annunciator input and inbuilt battery backup but may be ordered without battery backup (Model ILD15NB) if a separate power supply can be guaranteed to maintain the lift hearing

augmentation driver operational on loss of lift car mains power supply for a period of not less than 2 hours.

- 4 Each lift car 100V PA speaker and loop driver shall be connected to its respective station platform public address zone by a single pair 100 V speaker cable looped to the lift controller cabinet from the nearest available station platform PA speaker position by the Specialist Electrical Subcontractor.
- 5 Wire from the lift controller cabinet to the 100 V PA speaker and loop driver on the lift car and terminate at both ends.

7.5.16 HEARING AUGMENTATION

Provide Ampetronic ILD15BB or approved equivalent hearing augmentation for the lift car, and connect by suitable trailing cable to the lift control cabinet.

7.6 LANDINGS

7.6.1 LANDING ENTRANCES

- 1 Entrances shall be certified with a – / 60 / – fire resistance rating to type tested prototype details, in accordance to AS1735.11
- 2 The floor surface within the landing door jamb zone may slope slightly downwards away from the door sill; the jamb shall be made so as to accommodate a maximum slope of 20mm within the landing door jamb zone.
- 3 Frames shall be installed square to building grid, horizontally in line relative to the liftwell front wall and vertically in line throughout travel of lift.
- 4 Frame installed plumb within ± 1.5 mm over height of frame.
- 5 If the front enclosure wall is of masonry construction or if door frames are not “grout-free” install a grout guard between masonry opening and frame to enable grouting of frame from outside liftwell.
- 6 The landing entrances jambs and heads for the goods lifts shall be solidly concrete-grouted in to provide a robust surface capable of withstanding heavy abuse from trolleys and the like. Groutless frames are not acceptable.
- 7 Door panels shall be reinforced to produce straight flush rigid panel free of distortion marks or blemishes or weld marks.
- 8 Where possible liftwell access devices shall be provided in the door frame rather than on the landing door. Access devices shall be provided to code requirements.

7.6.2 LANDING DOORS

- 1 Landing doors shall be of the horizontal sliding type designed for use with power door operator.
- 2 Doors shall open to the specified clear opening width when open.

- 3 Doors shall be aligned with adjacent car and landing entrance jambs when open.
 - 4 Doors shall be constructed of furniture steel underlying construction suitably reinforced to achieve rigid assembly.
 - 5 Doors shall incorporate removable guides on bottom edge preventing lateral oscillation and noise.
 - 6 Movement and discernible noise shall be prevented while car is travelling.
-

7.6.3 LANDING SILLS

- 1 The landing sills shall be constructed of extruded hard aluminium alloy.
 - 2 Sills shall be installed flush with the finished floor level.
 - 3 Supply and fix all necessary support angles, brackets, bolts and hardware together with mild steel forming strips under sills to avoid slurry entering liftwells when sills are grouted in.
-

7.6.4 RECALL FIRE SERVICE CONTROL SWITCH

- 1 Fire recall control switch(s) shall be provided per BCA.
 - 2 They shall be located on the main floor call button station(s). Location as specified elsewhere
 - 3 Faceplates shall be surface engraved to requirements of Standard.
-

7.7 LIFT MACHINE EQUIPMENT

7.7.1 LIFT MACHINES

- 1 Permanent Magnet machines are preferred.
- 2 Gearless traction type machines shall consist of a motor, drive sheave and electro mechanical brake mounted on a common bed plate. The machine shall be designed, manufactured, assembled and balanced to ensure minimum vibration, noise or excessive wear.
- 3 The machine shall be designed, manufactured, assembled and balanced to ensure minimum vibration, noise or excessive wear.
- 4 The machine shall have isolation pads fitted to eliminate the transmission of sound and/or vibration.
- 5 Where integral fan cooling is provided to the hoist motors, the fan shall be thermostatically controlled or continue to run for a period of approximately 180 seconds after the motor has stopped.
- 6 All bearings shall be precision grade, self-aligning and quiet in operation.
- 7 The machine shall be provided with a nameplate setting out lift duty, motor speed, drive sheave diameter and size and gross lengths of hoist ropes for replacement purposes.

7.7.2 MRL CONTROLLER CABINETS

- 1 The following shall be provided in every controller cabinet in the case of MRL lifts:
 - a Distribution boards incorporating:
 - i Complete distribution board designed to withstand the anticipated fault level determined in conjunction with the Electrical services provider.
 - ii Complete distribution board designed to appropriate IP level.
 - iii Adequate labelling of circuit breakers.
 - b Locate adjacent to access point and incorporate terminations for installed submains and dedicated circuit breakers with lock off facility Including:
 - i Main switch for each lift.
 - ii Liftwell lighting and pit power with RCD.
 - iii Car light and power with RCD.
 - c One set of as-installed wiring diagrams in clear plastic envelopes suitably mounted.
 - d Each lift controller shall contain an engraved performance notice stating the guaranteed:
 - i Door open and close times.
 - ii Door dwell times.
 - iii Floor levelling accuracy.
 - iv Acceleration and deceleration rates.
 - v Jerk rate.
 - vi Contract speed.
 - vii Flight times (door open to door open) for one, two and four floor runs.
 - e Lighting and Power:
 - i All distribution boards and electrical connections shall be in accordance with the latest version of AS 3000.

7.7.3 DYNAMIC BALANCING OF MACHINES

- 1 After installation and setup, dynamically balance all rotating equipment in the machine room area for each lift including worm shafts, rotors, armatures, brake couplings and flywheels where applicable.

7.7.4 MACHINE AND HITCH BEAMS

- 1 The machine shall be mounted on structural steel beams supplied and installed with bearing plates or steel upstand.

- 2 Machines and/or machine beams shall be isolated from the machine beams and/or floor slabs by noise and vibration isolating mounts designed to suit the application.
- 3 Where 2:1 roping is used supply steel hitch beams, suitable steel bearing plates and isolating mounts for these beams.
- 4 The bearing plates and Unistrut inserts where required, shall be cast into masonry plinths or the structure as necessary.
- 5 Should machines be mounted directly on reinforced concrete upstands, provide structural engineering certification verifying the suitability of the upstands and floor slab to withstand the imposed dynamic loadings under all conditions.

7.8 LIFTWELL EQUIPMENT

7.8.1 GUIDE RAILS AND FIXINGS FOR ALL LIFTS

- 1 Guide rails shall be planed tee section. Deflections are to comply with the requirements of EN81--20
- 2 Extreme care shall be taken during storage, packing, handling and transporting of rails throughout the supply and erection process such that rail quality is maintained.
- 3 Sufficient consideration should also be given to the isolation of lift guide rails by which vibration may be transferred to the slabs

COUNTERWEIGHT

- 1 Counterweight guide rails shall be tee section.

GUIDE RAILS

- 1 Guide rails shall be installed plumb and parallel to ensure smooth riding quality.
- 2 Tee type guide rail joints shall be machined and factory matched tongue and groove type which employ fish plates of cross sectional area at least equivalent to guide rail cross section.
- 3 Rail clip design shall afford vertical rail movement during contraction, expansion, building settlement or safety gear application without distortion to rails. Any paint or corrosion treatment of rails should not interfere with this requirement.
- 4 Guide rails shall be either pinned below the lift well roof or bottomed on the pit frame with adequate provisions made for shortening of rails and/or adjusting jacking bolts to allow for building settlement.
- 5 Supply and install all trimmer beams, brackets, backings, stiffeners, packings, bolts and similar items necessary for the support and fixing of all guide rail systems.

7.8.2 WELL FLUSHING

- 1 Supply and install sheet steel fascia panels between the sill of each opening and the header plate of the opening below, above the top opening and below the bottom opening in the path of travel of the car door opening as required.
- 2 Flushing panels shall be suitably stiffened and painted prior to practical completion.
- 3 In scenarios where the distance between the liftwell face and the car sill is greater than 150mm or if the liftwell is glass, car door locking shall be provided.

7.8.3 SCREENS

- 1 Where required by code, supply and install pit, well and counterweight screens and any guarding necessary for the prevention of travelling cables being damaged by the shaft or lift equipment.

7.8.4 COUNTERWEIGHT

- 1 The counterweight shall be of structural frame and designed to be balanced as per code requirements.
- 2 Counterweight and associated equipment shall be designed and installed to minimise noise. Noise from the counterweight or associated equipment should not be audible from within the lift car, or on any of the occupiable areas. Any noise evident in any of the areas shall be rectified.
- 3 Counterweight shall be painted bright yellow for safety purposes. For scenic lifts, the counterweights are to have aesthetically designed covers.
- 4 Each counterweight shall be fitted with guide shoes, spring loaded and adjustable to ensure sufficient contact with the guide rails.
- 5 Prior to the top set of guide shoes being finally bolted in position, adjustment shall be made to ensure the counterweight will hang vertical at the mid-point of travel.

7.8.5 PIT ACCESS

- 1 Supply and install a suitable access ladder to allow access from the level of the lowest floor served by each lift to the lift pit. Extend the ladder a minimum of 1,100mm above the lowest floor level. Means shall be provided to allow unlocking of the landing doors from the pit floor.
- 2 Provide a safe means of access for pits greater than 2m deep.

7.8.6 PIT STOP SWITCH

- 1 Suitable impact resistant STOP switches shall be provided adjacent to the top and bottom of each pit access ladder and labelled. All STOP switches to be push / pull type.
- 2 Provide overspeed governor wheel tension or broken rope detection switches in each lift pit.

7.8.7 PIT AND LIFTWELL LIGHTING AND POWER

- 1 Fluorescent luminaires, with a suitable impact resistant diffuser shall be supplied and installed in each pit and liftwell. In addition, one RCD protected socket outlet shall be provided in each lift pit. The power supply to this outlet, is to be provided from a separate supply to the pit lighting. The pit lighting shall have emergency lighting facility.
- 2 Fluorescent luminaires, with impact resistant covers, shall be installed at every floor and provide a minimum light level of 50 lux at the car top wherever the car is located in the shaft. The top and bottom fittings are to be located 500mm from the top of the shaft and pit floor. The bottom light fitting of the shaft lighting in all shafts shall have an emergency light facility.
- 3 Liftwell luminaires are to have switching points from the car top, machine area and pit. In the case of multiple shafts the lights shall be switched from any car top or any pit entry point for the bank of lifts.
- 4 Two adjacent liftwells may share a single riser and three or four adjacent liftwells may share two risers where effective illumination level of 50 lux one metre above car roof can be achieved at any point of travel.

7.8.8 BUFFERS AND SUPPORTS

- 1 Provide buffers designed to the requirements of AS1735 in the bottom overrun pit.
- 2 Hydraulic buffers shall be fitted with graduated dip sticks to enable checking of the oil level, capped filler openings to facilitate topping up and suitable drain plugs. Oil reservoirs shall be sealed to prevent leakage and contamination.
- 3 Bolt buffers to mild steel bases fixed to the pit floor between the guide rails. Pit steel, bases, buffers and any braces required shall be supplied and installed.
- 4 Manufacture pedestals to allow buffers to be supported above the pit floor from steel sections. Pedestals shall be suitably plated at each end to allow adequate fixing of the buffer to the spacer and of the pedestal to the pit floor bases.
- 5 Where the vertical height from the pit floor exceeds 760 mm to the top of the compressed buffer, the top of the buffer shall be supported in a steel framework fixed at the ends to the guide rails.

7.8.9 LIFTWELL COMMUNICATIONS

- 1 An integrated communication system shall be provided that allows communication between the lift machine room, lift car, top of car and lift pit. Pixel or approved equivalent.

7.8.10 LIFT TRAVELING CABLE REQUIREMENTS

- 1 In addition to those cables specified two spare shielded twisted pair cables shall be included.
- 2 All cable cores shall be terminated on appropriate termination terminals to allow 3rd party connecting to be achieved, as close as possible to the lift machine room area entrance. The twisted pair cables and co-

axial cables shall be terminated in an accessible location to allow one access for the connection of a 3rd parties equipment, e.g., access control, CCTV, etc.

7.8.11 LIFTWELL EQUIPMENT FINISHING

- 1 Paint liftwell equipment in standard gloss enamel colour.
-

7.8.12 GOODS PASSENGER LIFT SIGNAGE

- 1 Goods passenger lift shall have a sign that is visible from the landing and loading area at all times, indicating the rated load capacity.
 - 2 The signage requirements of EN81-20 including clause 5.4.2.2.1 5) shall be met.
-

7.8.13 BUILDING TOLERANCE

- 1 Lifts shall be designed to be capable of being installed in a lift shaft built within the building tolerance required by the Contractor.

8 MONITORING

8.1 BUILDING MANAGEMENT SYSTEM

LIFTS

- 1 A relay based BMS interface for all lifts to monitor status and fault alarms shall be provided as follows:
 - a Signal 1 – Consolidated Fault, which would be activated when any one or more of the following conditions has occurred:
 - i Lift failed to start.
 - ii Lift power failure.
 - iii Lift turned off.
 - iv Lift out of service (maintenance).
 - v Lift safety device operated.
 - b Signal 2 – Alarm button pressed, which would be activated when the following condition has occurred:
 - i Car alarm button pressed for more than 3 seconds.
 - c Signal 3 – Fire Service
 - d Signal 4 – Exclusive Service
 - e Signal 5 – Water sensor in pit activated
-

8.2 REMOTE MONITORING BY A BUILDING MANAGEMENT SYSTEM (BMS)

- 1 Where specified, an interface to a management system is to be provided. It is to be located in the Security Room (or location designated by the client) and shall have the ability to monitor and every lift within the building. BMS system is to allow for the following:
 - a Monitoring:
 - i Car & Landing controls.
 - ii Fire Service.
 - iii Emergency Alarm.
 - iv Lift car parked or 'Out of Service'.
 - v Lift Machine Area over temperature alarm.
 - b Lift failed to start alarm.

Data is to be stored on electronic media and have the capability to be down loaded in both graphic and alpha numeric format to the printer or electronic media. All data shall be capable of being displayed on the screen in the formats described above and within time frames from a minimum of 5 minutes periods upwards.

8.3 CLOUD-BASED LIFT MONITORING

- 1 Provide a cloud-based lift monitoring system that consistently monitors the condition of the lift equipment, providing comprehensive monitoring and predictive maintenance capabilities.
- 2 The lift monitoring system shall include the following features:
 - a Real-time monitoring of lift equipment condition.
 - b Data storage and processing in a secure cloud environment.
 - c Predictive maintenance capabilities to anticipate part replacements.
 - d Remote access and control through a user-friendly interface.
- 3 The lift monitoring system shall consist of both hardware and software components, including but not limited to:
 - a Sensors and monitoring devices to collect data.
 - b A cloud-based server infrastructure for data storage and processing.
 - c User interface software for configuration and access.
 - d Connectivity options for seamless data transmission.
- 4 The System shall continuously collect data on the following parameters:
 - a Usage patterns.
 - b Lift equipment performance.
 - c Ride quality metrics.
- 5 The collected data shall be analysed in real-time, and the System shall use machine learning and predictive algorithms to determine when parts of the lift equipment require maintenance or replacement.

TENDER FORM

FOR

TRANSPORTATION SERVICES

AT

CQU ROCKHAMPTON TAFE CONSOLIDATION – STAGE 2

I/We, the undersigned, hereby tender to carry out the complete works in accordance with the Specification, Drawings, General Conditions of the proposed contract and addenda, for the sum set out hereunder:

Fixed Lump Sum Tender excluding Goods & Services Tax (GST) for the complete Transportation Service Works Package as specified:	\$
Provisional Sums:	\$
GST:	\$
TOTAL INCLUDING GST:	\$

ADDENDUM

We, the undersigned, acknowledge having received the following addenda and have incorporated them in our tender proposal:

NUMBER	DESCRIPTION	DATE RECEIVED

DATED THIS DAY OF 2026
SIGNATURE OF TENDERER
COMPANY
ADDRESS

T.1 Contract Maintenance after Final Completion

We, the undersigned, tender to carry out Contract Maintenance for Transportation Services in the above project in accordance with the Specification document and any noted Addenda if we are appointed for the supply and installation of the equipment.

Validity Period - we agree that this offer may be accepted by the building owner or agent at any time up to the expiration of the Defects Liability Period.

Total Tender Price for Transportation Services Comprehensive Maintenance after Final Completion subject to Rise and fall from date of tender, price per year, for the sums below.

Item	Whole dollars
Total fixed lump sum annual maintenance price – 3 years	\$
Total fixed lump sum annual maintenance price – 5 years	\$

T.2 Compliance Statements

The following compliance statements override any and all other statements and contents in the tender documents.

The Equipment provided in accordance with the Total Fixed Lump Sum Tender price for Transportation Services to Final Completion including Defects Liability Period Maintenance will be fully Complying as regards the following:

- 1 The Specification and Addenda including Specification Drawings (Yes/No) (attach full list of deviations if No)
- 2 Relevant Regulations and Acts (Yes/No) (attach full list of deviations if No)

T.3 Tender Price Schedule

The total fixed lump sum tender price excluding GST and excluding PC sums for the complete Transportation Service Works Package as specified **is the total of** the following prices excluding GST and excluding PC sums:

Item	Whole dollars
Building 33 Lift No. 1	\$
Building 36 Lift No. 1	\$
Building NTB Lift No. 1	\$
<i>Total fixed lump sum tender price as above excluding GST and excluding PC sums for the complete Transportation Service Works Package as specified</i>	\$

T.4 Major Tender Price Components

The total fixed lump sum tender price excluding GST and excluding PC sums for the complete Transportation Service Works Package as specified **is the total of** the following prices excluding GST and excluding PC sums:

Item	Whole dollars
Materials and manufactured equipment	\$
Imported equipment	\$
Site installation labour	\$
Maintenance between Practical completion and Final Completion	\$
Total fixed lump sum tender price as above excluding GST and excluding PC sums for the complete Transportation Service Works Package as specified	\$

T.5 Other Tender Price Components

The total fixed lump sum tender price excluding GST and excluding PC sums for the complete Transportation Service Works Package as specified **contains** the following prices excluding GST and excluding PC sums:

Item	Whole dollars
	\$
	\$
	\$
	\$
	\$

Signature of Tenderer: _____

Date: _____

T.6 Alternatives and Options

The following alternatives and/or options are offered as variations to the above total fixed lump sum tender price excluding GST and excluding PC sums for the complete Transportation Service Works Package as specified. Attach details where items are not fully specified.

Item	Total saving - whole dollars	Total extra - whole dollars
	\$	\$
	\$	\$
	\$	\$
	\$	\$
	\$	\$

T.7 Program

We propose the following as the earliest normal delivery and installation program, where “Week No.” means the start of that week number, in calendar weeks, counting in successive weeks from week No. 1, and where areas are weatherproof and suitable to take over, and subject to the necessary progressive completion of works to be performed by others:

LIFT

STAGE	Building 33 Lift No. 1	Building 36 Lift No. 1	Building NTB Lift No. 1
Receipt of written instruction to proceed	Week 1	Week 1	Week 1
Produce of shop drawings			
Approval of shop drawings			
Imported equipment ready to place on site			
Contractor's lift available (if req'd)			
Contractor's lift returned			
Practical Completion			

T.8 Defects Liability Maintenance

AVERAGE RESPONSE TIME TO SERVICE CALL OUT		GUARANTEED MAXIMUM RESPONSE TIME TO SERVICE CALL OUT		
During normal working hours	Outside normal working hours	During normal working hours	Outside normal working hours	Trapped passengers

..... minutes minutes minutes minutes minutes

T.9 Reference Installations

Recent installations where equipment of a similar design to that specified has been installed and completed.

LIFT

Project name	Installed Date	Location	Load / speed	No. floors
.....				
.....				
.....				
.....				

T.10 Proposed Subcontractors

Work proposed to be subcontracted by the tenderer:

Description of Work	Proposed Subcontractor	Reference Project where subcontractor has previously done this kind of work for the tenderer
.....		
.....		
.....		
.....		

T.11 Lift Equipment Description:

The lift equipment offered is as per the below:

PARAMETER	Building 33 Lift No. 1	Building 36 Lift No. 1	Building NTB Lift No. 1
Design Code			
Rated Load (kg)			
No. Passengers			
Speed (m/s)			

Signature of Tenderer: _____

Date: _____

PARAMETER	Building 33 Lift No. 1	Building 36 Lift No. 1	Building NTB Lift No. 1
Rating (Pass or Goods A etc)			
Car Width between applied finishes (mm)			
Car Depth between applied finishes (mm)			
Clear height to false ceiling			
Entrance Type			
Entrance Size (W x H)			
No. COP's per car			
Communication system type			
Button type			
Drive (G'd, G'less)			
Gearbox type if G'd			
Roping (1:1, 2:1)			
Motor rating (kW)			
Rated starts per hour			
Maximum Demand (A)			
Maximum current duration			
Heat output (kW)			
Controller manufacturer			
Power door operator model description			
Power door operator rated starts per hour			
Door jamb description			
Guide shoe type			
Rope brake manufacturer (if req'd)			
Car indicator display manufacturer / Type (LED, Dot Matrix)			

T.12 Deviations from Specification

.....

.....

Signature of Tenderer: _____

Date: _____

.....

.....

.....

.....

.....

.....

.....

.....

Signed for and on behalf of the tenderer

In signing below the signatory personally warrants that he/she has the actual authority to bind the Tenderer to this tender.

Tenderer (name of Company tendering)

Tenderer's ACN	Tenderer's ABN
.....	

Name of person signing for and on behalf of the tenderer

Office held by person signing for and on behalf of the tenderer

Signature

Date of signing

Signature of Tenderer: _____

Date: _____

APPENDIX A

TECHNICAL DATA

A1 TECHNICAL DATA

The following technical data form shall be submitted during the shop drawing phase of the project.
One form shall be completed for each group of lifts provided on the site.

Shop Drawing Technical Data

Lift Numbers:

CQU Rockhampton TAFE Consolidation – Stage 2

1. Lift Numbers:

2. Name and Address of VT Subcontractor:

3. Name and Address of the Owner/Principal:

4. Name and Address of the Contractor:

Duty Information:

1. Type of lift equipment (Conv Overhead or MRL):

2. Lift Classification (Pass or Pass/Goods Class ?)

3. Rated Load (Pass / kg):

4. Rated Speed:

5. Levels served:

6. Travel of lift:

Machine and Controller Information:

1. Machine type (geared or gearless):

2. Machine size (kW):

3. Machine reference model number:

4. Control type (Conv or Destination Control)

5. Controller Model number:

6. Drive type (VVVF, VFAC, etc):

7. Regen drive:

Governor Information:

Shop Drawing Technical Data

Lift Numbers:

CQU Rockhampton TAFE Consolidation – Stage 2

1. Governor model number:

1. Governor mechanical tripping speed:

2. Governor electrical tripping speed:

3. Rope diameter:

4. Rope type:

5. Rope construction:

6. Breaking load:

Car Information:

1. Car guide rail size and type:

2. Car roller guide type and size:

3. Safety gear type:

4. Safety gear model number:

5. Roping ratio (i.e. 2:1)

6. No. hoisting ropes:

7. Size of hoisting ropes:

8. Hoisting rope type:

9. Hoisting rope construction:

10. Hoisting rope breaking load:

11. Compensation (rope or chain):

12. Compensation type (lockdown or non-lockdown):

13. Compensation size and number of ropes:

14. Compensation rope type and construction:

15. Compensation rope breaking load:

16. Mass of car (kg):

Counterweight Information:

1. Counterweight guide rail size and type:

2. Counterweight roller guide type and size:

3. Counterweight safety gear type:

Shop Drawing Technical Data

Lift Numbers:

CQU Rockhampton TAFE Consolidation – Stage 2

4. Counterweight safety gear model number:

5. Mass of the counterweight:

Buffer Information:

1. Car buffer type (Spring, oil, neoprene):

2. Car buffer model number:

3. No. of car buffers:

4. Counterweight buffer type (Spring, oil, neoprene):

5. Counterweight buffer model number:

6. No. of counterweight buffers:

Door Operator Information:

1. Door operator model number:

Electrical Information:

1. No. submain supplies:

2. Submain circuit breaker size:

3. Electrical Maximum Demand (total):

4. Running current per lift:

5. Starting current per lift:

6. Fault level:

7. Heat liberation (total):

Commissioning Data:

1. Emergency phone directed to (lift manufacturer or on site location):

2. Governor mechanical tripping speed:

3. Governor electrical tripping speed:

4. Safety gear test date (pass or fail):

5. Fire service operation successful: